

The Evolution of Managerial Expertise: How Corporate Culture Can Run Amok

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This paper investigates how noisy evaluation of worker skills affects human capital investments and hiring. Individuals distort investments toward skills that most managers can evaluate. Dynamically, when workers become managers, managerial expertise can become increasingly skewed over time, raising investment distortions and reducing output. If firms select managerial expertise strategically, efficient investments can be retrieved when (a) identifying whether workers' skills matter more than distinguishing among skilled workers, and (b) initial investment distortions are small. Otherwise, such strategic design worsens long-run outcomes. Finally, we determine when short-run affirmative action policies are effective. (JEL D20, J24, J41)

Suppose that employers can better evaluate some attributes of job applicants than others. One reason may be that certain attributes are easier to evaluate. Alternatively, those making hiring decisions, having themselves been hired because they possess certain attributes, can better evaluate the same attributes in others. If job applicants know this, they may skew skill development toward those skills that employers can best evaluate, even if those skills contribute less to on-the-job productivity than others. In turn, in dynamic contexts, where workers eventually become managers, the distribution of managerial evaluation expertise can grow increasingly distorted over time, reducing output.

The consequences of the fact that recruiters may be better able to evaluate some skills than others are manifest in professional sports teams. In many team sports, goals and assists provide tangible evidence that makes offensive skills

easier to evaluate than defensive skills, teamwork, and other intangibles. To increase their chances of being drafted, young players devote most of their time toward improving those skills that are more easily evaluated, focusing on offensive skills. From a dynamic perspective, scouts, who are typically former players, emphasize offensive skills in their training, and hence are able to evaluate better these same skills, reinforcing investment distortions.

This paper investigates the consequences of such noisy skill evaluation for skill investment and hiring in a simple overlapping generations framework. We consider two types of job applicants, with each type having a comparative advantage in developing one of two skills. Evaluation is noisy because each manager can evaluate only one of the skills. Job applicants invest in skill development, taking this noisy evaluation procedure into account. Those hired become the next crop of managers and make the next period's hiring decisions. We then derive how the determinants of a manager's evaluation expertise affect the evolution of skill investments and worker productivity.

We first analyze a potential employee's skill investment problem *given* a particular managerial expertise distribution. We show that all agents skew investments toward the skill that most managers can evaluate. This is not socially efficient *even* when most managers can evaluate the more productive skill. Output is reduced

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both because of the decreasing marginal returns to investment in a given skill, and because those with innate talents in developing the less productive skill inefficiently shift investments toward the other skill. Corporate culture appears to run amok, as managers seem to hire too many of their own type. Dynamically, if the expertise of managers depends sufficiently on the human capital investments that they make, then managerial evaluation skills can become increasingly skewed over time, further raising investment distortions and reducing aggregate output. In turn, the associated long-run stable steady states feature distributions of managerial expertise in the economy that are more unbalanced than is optimal.

In our base model, a firm's managerial expertise is determined solely by past hiring decisions, and managers hire to maximize period profits. One can imagine, though, that firms might design compensation or personnel departments more strategically. Accordingly, we determine how outcomes are affected if some firms *choose* the expertise of their managers. Compensating managers with equity rather than claims to period profits has similar effects: equity leads managers to set lower productivity standards for workers who are more likely to have the "right" expertise.

We show that these forms of strategic behavior retrieve efficient human capital investments in the long run when (a) identifying whether workers are skilled matters more than distinguishing among skilled workers, and (b) the initial economy-wide distribution of managerial expertise is not too unbalanced so that worker investments are not too distorted. Under such conditions, firms do better to investigate the skill that few managers in the economy can evaluate. Importantly, in this case, strategic firms can make an expertise-balanced steady state locally stable.

If these two conditions do not hold, however, strategic consideration of managerial expertise, while privately optimal, is socially suboptimal, increasing investment distortions and reducing long-run output in the economy. In particular, if there is enough dispersion in realized skills, so that identifying the *level* of a worker's skill contributes more to profits than identifying more *multiskilled* workers, firms would do best to select managers who can evaluate the skill

that most workers acquire. So, too, if the economy-wide distribution of managerial expertise, and hence human capital investments, is too skewed, a firm should again evaluate the predominant skill. In such scenarios, strategic selection of the predominant managerial expertise skews further the distribution of managerial expertise in the economy, reinforcing investment distortions toward the skill that most managers can evaluate, speeding the economy's convergence to inefficient steady states.

Our findings have implications for the efficacy of temporary affirmative action policies. For example, in an economy where men and women tend to be better at developing different skills, a historical bias toward hiring men can be entrenched, as yesterday's male hires are today's evaluators. A policy that mandates hiring more women raises the number of managers who can evaluate the skills that most women develop. The more balanced distribution of managerial expertise generated by affirmative action reduces investment distortions. In turn, when firms compensate managers with equity, if identifying more multiskilled workers contributes more to profits than identifying the level of a worker's skill, then the temporary affirmative action policy will "take," as profit-maximizing firms would *choose* to hire more women. This leads to an even more balanced distribution of managerial expertise and investments, and the affirmative action policy becomes self-enforcing.

The underlying logic of our model is related to that in Bengt Holmström and Paul Milgrom's (1991) multi-tasking model, in which agents distort effort toward dimensions of effort that an employer can observe. Our paper develops the dynamic and general equilibrium consequences of this insight. Our paper is also related to work by Bradford Cornell and Ivo Welch (1996), and Susan Athey et al. (2000). Using a model similar to ours, Cornell and Welch (1996) show that noisy evaluation can lead to hiring discrimination even if employers do not intend to discriminate. Stephen Coate and Glenn C. Loury (1993), Andrea Moro and Peter Norman (2003), and Norman (2003) show that multiple equilibria can arise in static contexts when there is noisy evaluation and workers invest in skills. In these papers, output depends on a worker's costly, unobserved human capital investment

and job placement (skilled or unskilled): only trained workers are productive in skilled jobs. Firms see a worker's sex and a noisy signal of whether the investment was made. Although sex does not directly affect investment costs, if firms believe, for example, that women are less likely to invest than men, then firms are more likely to place women in unskilled jobs, making investment by women less attractive. As a result, these beliefs can be self-fulfilling. The seminal paper is by Edmund Phelps. Phelps (1972) supposes that firms have noisier signals of some agents' abilities than others, and derives the static implications. In some sense, the subsequent literature serves to endogenize the noise in signals and the consequences for skill investment. In addition, our paper contributes by deriving the dynamic and output implications.

Athey et al. (2000) provide a model of promotional policies that seeks to explain the evolution of diversity within a firm. In their paper, entry-level employees receive mentoring only from managers who are of their "type." This mentoring augments the ability of lower-level employees, increasing their chances of being promoted. Consequently, while entering workers in a firm may be diverse, higher echelons are not. In a similar vein, Canice Prendergast (1993) shows how subjective performance evaluation can give rise to "yes men," workers acting to conform to the opinions of their supervisors. Our analysis extends this research agenda in that we first investigate how the distribution of managerial types affects skill investments by potential employees, and we then derive the consequences for the evolution of the distribution of managerial types over time.

Evidence of the effects of noisy evaluation processes can be found in David Granick's (1972) study of managerial selection across four countries. Granick found that the probability that a young manager is promoted is determined in large part by whether his skills coincide with those of the superior. In a study of large and medium-sized U.S. firms, Michael Useem (1989) found that in many companies the proportion of managers with one skill or another was exceptionally high or low. For example, in some companies the "MBA culture" dominates, as "[companies] want their own type. The frame of reference here is the MBA" (Useem, p. 125).

Useem finds that if a CEO holds an MBA degree, the chief operating officer is 2.06 times as likely to have an MBA.

Lastly, our paper complements the literature on uniform social behavior (e.g., Thomas Schelling, 1978). These models generate local conformity through coordination externalities in an adopted technology (e.g., Brian Arthur, 1989, 1994), herding through reputational effects (e.g., David Scharfstein and Jeremy Stein, 1990; Jeffrey Zweibel, 1995), or learning (Abhijit Banerjee, 1992; Sushil Bikhchandani et al., 1992). Here, it is the incentive to get hired, coupled with noisy evaluation of job applicants, that leads to "conformity" in investments.

The paper is organized as follows: Sections I and II present the model, analysis, and findings. Section III quantifies our findings numerically and illustrates how outcomes are affected when core assumptions are altered. Section IV explores the consequences of strategic design of personnel departments, or compensating managers with equity. Section V concludes. Proofs are in the Appendix.

I. The Model

A. Overview

In each period of our overlapping-generations model, a continuum of measure one of risk-neutral, three-period-lived agents is born. In their first period of life, agents are endowed with one unit of time that they can devote to developing two skills, *A* and *B*. In an agent's second period of life, these skills are productive if and only if the agent is employed as a white-collar worker. In their third period of life, white-collar workers become managers. Managers develop expertise in evaluating one of the two skills and determine which job applicants to hire.

B. Skill Development

Fraction $\beta_A \in (0, 1)$ of young agents are better able to develop skill *A*, while fraction $\beta_B = 1 - \beta_A$ are better able to develop skill *B*. Specifically, a type-*i* agent who devotes time t_i to developing skill *i* and time $1 - t_i$ to the other skill *j* acquires skill *i* with probability $H(t_i)^\psi$,

and acquires skill $j \neq i$ with lesser probability $L(1 - t_i)^\psi$, where $0 < L < H < 1$ and $0 < \psi < 1$. There is no aggregate uncertainty in the economy, so that, for example, if all type- i agents devote time t_i to developing skill i , then a measure $\beta_i H(t_i)^\psi$ of them acquire skill i . As in Cornell and Welch (1996), we assume that the realized skill level, $s_k > 0$, of an agent who succeeds in developing skill k is drawn from a distribution $F(\cdot)$, with associated positive density, $f(\cdot)$, on its bounded support, $[\underline{z}, \bar{z}]$, with mean $\bar{s} = E(s_k | s_k > 0)$. An agent who does not acquire skill k has $s_k = 0$.

The output of a white-collar worker who acquires skill levels s_A and s_B is

$$(1) \quad Y = e_A s_A + e_B s_B,$$

where $e_k > 0$ reflects the contribution of skill k to output. Without loss of generality, we assume that $e_A \geq e_B$, so that skill A is more valuable, and we normalize the price of a unit of output to 1.

C. Managerial Expertise

Upon becoming a manager, a worker develops an evaluation expertise in *one* of the two skills. In particular, a manager with type- k expertise *perfectly* observes an applicant's level of skill k , but *cannot* observe either the other skill or the applicant's type. To capture different possible determinants of managerial expertise, we let the transition from a type- i worker to a type- k manager reflect both the worker's innate ability and the time invested in acquiring skill k . Specifically, a type- i worker becomes a manager with expertise in skill i with probability $(1 - \gamma) + \gamma t_i$, and develops expertise in the other skill j with probability $\gamma(1 - t_i)$. That is, $\gamma \in [0, 1]$ captures the importance of "nurture" (time invested) relative to "nature" (innate ability) for determining managerial expertise. The higher is γ , the more important is nurture.

D. Firms

There are N firms in the economy. Each firm has job openings for a measure ν/N of white-collar workers. At date 0, at each firm, fraction μ_0 of managers have expertise in skill A . Each young agent can apply to only one firm and is

evaluated by a single manager at that firm. The probability that the young agent is evaluated by a manager with expertise in skill A is equal to the fraction of managers at the firm with expertise in skill A .

E. Compensation

Agents who are not hired to white-collar positions pursue a less attractive occupation with a discounted lifetime wage that we normalize to zero. White-collar workers receive a common wage $W \geq 0$ which does not exceed the expected output of the marginal worker hired. Managers at a firm receive equal shares of the period profits derived from their collective hiring decisions.

The assumption that entry-level workers receive the same wage is consistent with standard practice in many organizations and, as we show, is unimportant for our findings.¹ In Section IV, we show how outcomes are affected if, instead of receiving shares of period profits, managers *sell* the firm to the workers at a price equal to the discounted value of profits.

F. Representative Firms and Agents

Because the date-0 distribution of managerial expertise, μ_0 , is the same at each firm, it is part of an equilibrium best response for the same measure β_k/N of type- k agents to apply to each firm. Then, with no aggregate uncertainty, at any date τ , at each firm, the same endogenous fraction μ_τ of managers has expertise in skill A . Hence, we can consider a representative firm with measure $\nu < 1$ of slots to fill out of a measure one of job applicants, where fraction μ_τ of its managers has expertise in skill A . So, too, each type- k agent makes the same investments, so that we can consider representative type- A and type- B young agents.

It helps to anticipate the equilibrium result that agents invest to maximize the probability that they are hired, while firms make hiring decisions to maximize period profits. With this

¹ In Section IV, we show that equilibrium investments and long-run outcomes are qualitatively unaffected if workers are paid their expected marginal product rather than a flat wage.

insight, we can characterize equilibrium strategies as a sequence of period strategies:

- A period strategy for young agents at time τ is a pair of functions mapping μ_τ into skill investments, $t(\mu_\tau) \equiv (t_A(\mu_\tau), t_B(\mu_\tau))$.
- A period strategy for the representative firm at time τ can be summarized by a pair of functions mapping μ_τ into hiring standards $\mathcal{K}(\mu_\tau) \equiv (\mathcal{K}_A(\mu_\tau), \mathcal{K}_B(\mu_\tau))$, where a manager with expertise in evaluating skill j hires a worker if and only if $s_j \geq \mathcal{K}_j(\mu_\tau)$.

Before we define an equilibrium, it helps to define some notation. For skill $j = A, B$, let $\mathcal{P}_j(\mu_\tau) \equiv 1 - F(\mathcal{K}_j(\mu_\tau))$ be the probability that a manager with expertise in skill j hires a job applicant given that the applicant has acquired skill j . Let

$$(2) \quad \Phi_A(\mu_\tau, t(\mu_\tau), \mathcal{K}(\mu_\tau)) \\ \equiv \mu_\tau \mathcal{P}_A(\mu_\tau) H(t_A(\mu_\tau))^\psi \\ + (1 - \mu_\tau) \mathcal{P}_B(\mu_\tau) L(1 - t_A(\mu_\tau))^\psi$$

be the probability that a type-A agent acquires a job, and define $\Phi_B(\mu_\tau, t(\mu_\tau), \mathcal{K}(\mu_\tau))$ analogously. Let the measure of agents that acquire both skills be

$$(3) \quad \sigma_{AB}(t(\mu_\tau)) \\ \equiv [\beta_A(t_A(\mu_\tau))^\psi (1 - t_A(\mu_\tau))^\psi \\ + \beta_B(t_B(\mu_\tau))^\psi (1 - t_B(\mu_\tau))^\psi] HL,$$

let the measure of agents that acquire skill A only be

$$(4) \quad \sigma_A(t(\mu_\tau)) \equiv \beta_A H(t_A(\mu_\tau))^\psi (1 - L(1 - t_A(\mu_\tau))^\psi) \\ + \beta_B L(1 - t_B(\mu_\tau))^\psi (1 - H t_B(\mu_\tau)^\psi),$$

and let the measure that acquires skill B only be $\sigma_B(t(\mu_\tau))$. From the perspective of a manager with expertise in skill i , the expected productivity of an applicant who develops skill i at date τ is

$$(5) \quad E[Y|s_i; t(\mu_\tau)] = e_i s_i + e_j E[s_j|s_i; t(\mu_\tau)] \\ = e_i s_i + \left(\frac{\sigma_{AB}(t(\mu_\tau))}{\sigma_i(t(\mu_\tau)) + \sigma_{AB}(t(\mu_\tau))} \right) e_j \bar{s}.$$

G. Equilibrium

Given an initial distribution of managerial expertise, μ_0 , a (Bayesian) equilibrium is a sequence of skill investment decisions, $\{t^*(\mu_\tau)\}_{\tau=0}^\infty$, and a sequence of managerial hiring standards, $\{\mathcal{K}^*(\mu_\tau)\}_{\tau=0}^\infty$, such that at each date τ :

- (a) *Agents optimize*: Skill investment $t^*(\mu_\tau)$, $i = A, B$, maximizes expected lifetime compensation,

$$(6) \quad \Phi_i(\mu_\tau, t(\mu_\tau), \mathcal{K}(\mu_\tau)) \\ \times \left(W + \delta \frac{\Pi^*(\mu_{\tau+1}, t^*(\mu_{\tau+1}))}{\nu} \right),$$

where δ is the discount factor and $\Pi^*(\mu_{\tau+1}, t^*(\mu_{\tau+1}))$ is the representative firm's equilibrium profit at date $\tau + 1$.

- (b) *Managers optimize*: $\mathcal{K}_A^*(\mu_\tau)$ and $\mathcal{K}_B^*(\mu_\tau)$ maximize period profits

$$(7) \quad \mu_\tau [\beta_A H(t_A^*(\mu_\tau))^\psi + \beta_B L(1 - t_B^*(\mu_\tau))^\psi] \\ \times \int_{\mathcal{K}_A(\mu_\tau)}^{\bar{z}} E[Y|s_A, t^*(\mu_\tau)] f(s_A) ds_A \\ + (1 - \mu_\tau) [\beta_A L(1 - t_A^*(\mu_\tau))^\psi \\ + \beta_B H(t_B^*(\mu_\tau))^\psi] \\ \times \int_{\mathcal{K}_B(\mu_\tau)}^{\bar{z}} E[Y|s_B, t^*(\mu_\tau)] f(s_B) ds_B \\ - W\nu,$$

such that the measure of workers hired at each date is ν , i.e.,

$$(8) \quad \nu = \beta_A \Phi_A(\mu_\tau, t(\mu_\tau), \mathcal{K}(\mu_\tau)) \\ + \beta_B \Phi_B(\mu_\tau, t(\mu_\tau), \mathcal{K}(\mu_\tau)).$$

- (c) *Evolution of μ_τ* : The fraction of managers with expertise in skill A evolves according to

(9)

$$\mu_{\tau+1}(\mu_\tau) = \frac{1}{\nu} [\beta_A((1 - \gamma) + \gamma t_A^*(\mu_\tau)) \\ \times \Phi_A(\mu_\tau, t^*(\mu_\tau), \mathcal{K}^*(\mu_\tau)) + \beta_B \gamma \\ \times (1 - t_B^*(\mu_\tau)) \Phi_B(\mu_\tau, t^*(\mu_\tau), \mathcal{K}^*(\mu_\tau))].$$

H. Firm's Problem

Because managers receive the profit generated by their hires, the firm hires to maximize next period's profits. Observe from equation (7) that only a worker's expected productivity, and not his likely managerial expertise, affects these profits. We will assume that the measure of white-collar positions, ν , is small enough that in equilibrium, $\mathcal{K}_j(\mu_\tau) > 0$, i.e., it is not optimal to hire an unskilled agent with $s_j = 0$. A sufficient condition for this is Assumption (A1):²

(A1) Let $\bar{\beta} = \max\{\beta_A, \beta_B\}$. Then ν is sufficiently small that

$$(10) \quad F^{-1}\left(1 - \frac{\nu}{\bar{\beta}L + (1 - \bar{\beta})H}\right) \\ \geq (\bar{\beta}L + (1 - \bar{\beta})H)\bar{s}.$$

Optimization by the representative firm then implies that an applicant is hired if and only if his expected productivity exceeds a cutoff, $\mathcal{K}(\mu_\tau)$, regardless of who evaluates him. $\mathcal{K}(\mu_\tau)$

is such that the measure of job applicants hired is ν . In particular, optimization by the firm implies that the expected productivity of the marginal hires is equated, so that

$$(11) \quad \mathcal{K}_A^*(\mu_\tau) = \frac{1}{e_A} \left[K(\mu_\tau) - \left(\frac{\sigma_{AB}}{\sigma_A + \sigma_{AB}} \right) e_B \bar{s} \right]$$

and

$$(12) \quad \mathcal{K}_B^*(\mu_\tau) = \frac{1}{e_B} \left[K(\mu_\tau) - \left(\frac{\sigma_{AB}}{\sigma_B + \sigma_{AB}} \right) e_A \bar{s} \right].$$

The firm's problem becomes more complicated if, rather than receive the profits generated by their hires, managers were to sell the firm to the current young workers. In this case, we show in Section IV that managers hire to maximize the discounted value of profits, or firm value. As a result, one cannot characterize hiring by a series of period strategies; the future evolution of μ_τ enters hiring decisions at date $\tau' < \tau$. In particular, the firm internalizes the implications of a worker's likely managerial expertise for future hiring decisions.

I. Agent's Problem

Observe from equation (6) that a type- i applicant chooses $t_i(\mu_\tau)$ to maximize his probability of being hired, $\Phi_i(\mu_\tau, t(\mu_\tau), \mathcal{K}(\mu_\tau))$. The two first-order conditions characterizing the two types' optimal investments are

$$(13) \quad \mu_\tau \mathcal{P}_A^*(\mu_\tau) H t_A^{\psi-1} \\ - (1 - \mu_\tau) \mathcal{P}_B^*(\mu_\tau) L (1 - t_A)^{\psi-1} = 0$$

and

$$(14) \quad (1 - \mu_\tau) \mathcal{P}_B^*(\mu_\tau) H t_B^{\psi-1} \\ - \mu_\tau \mathcal{P}_A^*(\mu_\tau) L (1 - t_B)^{\psi-1} = 0.$$

Solving the first-order conditions yields³

² If almost all managers evaluate the skill k that fewer young agents have an innate ability to acquire then measure $m = \bar{\beta}L + (1 - \bar{\beta})H$ of young agents develop skill k and fraction ν/m are hired. Hence, the marginal worker hired has $s_k = F^{-1}[1 - (\nu/m)]$. (A1) says that an agent who is known not to have developed the skill that most agents are better at.

³ Note that the solutions for t_A and t_B do not depend on agents being risk-neutral, extending to any indirect utility functions $U(m_1, m_2)$ where m_1 is income as a worker, m_2 is income as a manager, and $U(m_1, m_2)$ is monotone in its arguments. This is because a job applicant's investments do

(15)

$$t_A^*(\mu_\tau) = \frac{(\mu_\tau \mathcal{P}_A^*(\mu_\tau)H)^{1/(1-\psi)}}{[(\mu_\tau \mathcal{P}_A^*(\mu_\tau)H)^{1/(1-\psi)} + ((1-\mu_\tau)\mathcal{P}_B^*(\mu_\tau)L)^{1/(1-\psi)}]}$$

(16)

$$t_B^*(\mu_\tau) = \frac{((1-\mu_\tau)\mathcal{P}_B^*(\mu_\tau)H)^{1/(1-\psi)}}{[(1-\mu_\tau)\mathcal{P}_B^*(\mu_\tau)H)^{1/(1-\psi)} + (\mu_\tau \mathcal{P}_A^*(\mu_\tau)L)^{1/(1-\psi)}]}$$

We first characterize how the time each type of agent devotes to acquiring each skill depends on the distribution of managerial expertise μ . Most importantly, we derive how μ_τ affects the hiring standards set by each managerial type and the probability a type- i agent is hired.

LEMMA 1: *Period Equilibrium Characterization.*⁴

- (a) *If no managers can evaluate a particular skill, agents optimally spend all their time developing the other skill: $t_A(0) = 0$ and $t_B(0) = 1$; $t_A(1) = 1$, and $t_B(1) = 0$.*
- (b) *Agents invest relatively more time in the skill in which they have a comparative advantage: i.e., $t_A(\mu_\tau) \geq 1 - t_B(\mu_\tau)$, with strict inequality if $\mu_\tau \in (0, 1)$.*
- (c) *The time invested by agents in skill A rises with μ_τ : $[dt_A(\mu_\tau)/d\mu_\tau] > 0 > [dt_B(\mu_\tau)/d\mu_\tau]$.*
- (d) *$\mathcal{P}_A(\mu_\tau)$ falls with μ_τ and $\mathcal{P}_B(\mu_\tau)$ rises with μ_τ . But $\mu_\tau \mathcal{P}_A(\mu_\tau)/[(1-\mu_\tau)\mathcal{P}_B(\mu_\tau)]$ is increasing in μ_τ , so that the chance of being hired with skill A increases with μ_τ .*

The key results are that raising the probability that a worker is evaluated by a type-A manager raises not only the time agents spend developing skill A, but also the measure of applicants hired by type-A managers, and the probability an agent with innate ability at developing skill A is hired. Part (c) follows because as μ_τ increases,

applicants who acquire skill A are better matched with evaluators, so they are more likely to be hired than is an applicant with skill B. This, in turn, provides an incentive for all agents to shift investments toward skill A. To understand part (d), first recognize that the more time agents devote to acquiring skill A, the less time they have for skill B. Hence, as μ_τ increases, the probability an agent who acquires skill A also acquires skill B falls; but the probability an agent who acquires skill B also acquires skill A rises. As a result, as μ_τ increases, type-A managers raise their hiring standard $\mathcal{K}_A^*(\mu_\tau)$, while type-B managers lower theirs; i.e., $\mathcal{P}_A(\mu_\tau)/\mathcal{P}_B(\mu_\tau)$ falls in μ_τ . This is more than offset, however, by the increasing proportion of type-As in the managerial pool. Taken together, the results of Lemma 1 indicate that regardless of innate ability, job applicants tend to skew skill investments toward those skills on which they are more likely to be evaluated.

Finally, observe that part (d) implies that hiring standards in a given skill rise with the proportion of managers who evaluate that skill. This implies that Assumption 1 is sufficient to ensure that a manager hires only workers who succeed in acquiring the skill that the manager can evaluate.

II. Steady State

In this section, we analyze the evolution of the distribution of managerial expertise given in equation (9) and determine the properties of the limiting distributions. We first show that given any initial distribution of managerial types, the distribution of managerial types always reaches a steady state in which the equilibrium distribution of managerial types remains unchanged from one period to the next, i.e., $\mu_{\tau+1}(\mu_\tau) = \mu_\tau$. Since $\mu_{\tau+1}$ maps the unit interval $[0, 1]$ onto itself, existence of a steady state follows from standard fixed point arguments as $\mu_{\tau+1}$ is continuous in μ_τ .

PROPOSITION 1: *There exists at least one steady-state distribution of managerial skills.*

We next characterize the steady-state distributions of managerial skills. A key question is whether the initial distribution of managerial expertise affects the steady-state distributions.

not affect his compensation conditional on being hired. We will show qualitatively how matters are affected if, instead, a worker's wage rises with his expected productivity.

⁴ Uniqueness of the period equilibrium follows because the time agents invest in skill A is increasing in $\mathcal{P}_A/\mathcal{P}_B$, while $\mathcal{P}_A/\mathcal{P}_B$ is decreasing in the time agents invest in skill A (see equations (11) and (12)).

We show that if and only if managerial expertise depends sufficiently on skill investment (nurture), then the steady state distribution depends critically on the initial distribution of expertise.

We first characterize equilibrium steady states for a symmetric environment in which the two skills contribute equally to output, and innate ability is equally divided between skills. Proposition 2 below reveals that if the evolution of managerial expertise is determined primarily by ability, then the balanced $\mu = 0.5$ mix of managerial expertise is the unique steady state. However, for all γ sufficiently large, multiple steady states arise as an unbalanced distribution of managerial expertise is reinforced by the unbalanced skill investments that agents make. In particular, the proposition implies that if nurture is a significant determinant of managerial expertise and the initial distribution of managerial expertise is sufficiently unbalanced, then the distribution of managerial expertise remains unbalanced forever.

PROPOSITION 2: *Consider any symmetric environment with $e_A = e_B$ and $\beta_A = \beta_B = 0.5$. Then*

- (a) *The balanced distribution of managerial expertise, $\mu = 0.5$, is always a steady state.*
- (b) *There exists a $\hat{\gamma} \in (0, 1)$ such that for all $\gamma < \hat{\gamma}$, the unique steady state is $\mu = 0.5$.*
- (c) *For all $\gamma > \hat{\gamma}$ there are multiple steady states, and the most extreme steady states, $\mu_+(\gamma)$ and $\mu_-(\gamma)$, are stable and symmetric, in the sense that $\mu_+(\gamma) > 0.5$ and $\mu_-(\gamma) = 1 - \mu_+(\gamma) < 0.5$.*
- (d) *As γ rises past $\hat{\gamma}$, the extreme steady states grow increasingly unbalanced: $\mu_+(\gamma)$ is strictly increasing in γ , and if nurture alone determines managerial expertise, the extreme steady states feature exclusively one managerial type or the other, i.e., $\mu_+(1) = 1$ and $\mu_-(1) = 0$.*

The Appendix contains both the proof to this proposition and to the claims below. We first prove very generally (i.e., even in asymmetric settings where one skill is more productive than another) that when nature alone determines managerial expertise, i.e., $\gamma = 0$, then $\mu_{\tau+1}(\mu_\tau) < 1$. It follows that there is a unique

steady state, and given any initial distribution of managerial expertise, there is monotonic convergence to the unique steady state. Even beginning with the most extreme distribution of managerial expertise, $\mu_0 = 1$, so that agents invest solely in skill A, some type-B agents acquire skill A and are hired, so that $\mu_1 < 1$. At date 1 this leads agents to invest some in skill B, more type-B agents are hired, and the process continues, converging to $\mu_\infty = 0.5$.

We then consider the other extreme, and suppose that nurture drives the evolution of managerial expertise, i.e., $\gamma = 1$. When $\gamma = 1$, then even in asymmetric settings, there are extreme steady states featuring exclusively one managerial type or the other, $\mu_+(1) = 1$ and $\mu_-(1) = 0$. This follows immediately from the observation that if all managers evaluate the same skill, then agents invest only in that skill, in which case future managers all possess that same expertise. The Inada feature of the skill acquisition technology implies that as $\mu_\tau \rightarrow 1$, both $dt_A(\mu_\tau)/d\mu_\tau$ and $-[dt_B(\mu_\tau)/\mu_\tau]$ go to zero, so that $\lim_{\mu_\tau \rightarrow 1} [d\mu_{\tau+1}(\mu_\tau)/d\mu_\tau] \rightarrow 0$, i.e., these extreme steady states are stable.

We then tie these observations together by deriving the impact of a marginal increase in γ on steady states. Define $D(\mu_\tau, \gamma)$ to be the difference between the measure of type-B employees who become type-A managers and the measure of type-A employees who become type-B managers.

$$(17) \quad D(\mu_\tau, \gamma) = \gamma[\beta_B(1 - t_B(\mu_\tau))\Phi_B(\cdot) - \beta_A(1 - t_A(\mu_\tau))\Phi_A(\cdot)].$$

The first term of $D(\mu_\tau, \gamma)$ rises with μ_τ and the second term falls, so there is a unique cross-over distribution of managerial skills, μ^T , such that $D(\mu^T, \gamma) = 0$. Because γ enters $D(\mu_\tau, \gamma)$ as a scalar, μ^T does not depend on γ —the cross-over distribution of managerial skills is not affected by the transition process for managerial expertise. In symmetric settings, $\mu^T = 0.5$. In asymmetric settings, generically, $\mu^T \neq 0.5$ and μ^T is not a steady state. In the Appendix, we show that at a steady state,

$$(18) \quad \frac{\partial \mu_{\tau+1}(\mu_\tau)}{\partial \gamma} = \frac{D(\mu_\tau, \gamma)}{\nu \gamma}.$$

Intuitively, γ does not affect agent investments, but if $\mu^T < \mu_\tau$, raising γ serves to raise the measure of managers with expertise in skill A in the next period, and if $\mu^T > \mu_\tau$, raising γ lowers the measure of expertise in skill A in period $\tau + 1$. It follows that the long-run steady states behave in exactly the same way. In particular, it follows that as γ is increased past $\hat{\gamma}$, these extreme steady states not only exist, but also grow increasingly unbalanced.

The last step is to extend this characterization to derive the impact of differences in skill productivities on long-run outcomes. We prove that $\mu_{\tau+1}$ is increasing in the skill productivity difference ($e_A - e_B$) for $\mu_\tau \in (0, 1)$, while μ^T is decreasing in ($e_A - e_B$). In particular, this implies that interior steady states are increasing in the skill productivity difference ($e_A - e_B$).

Finally, in the Appendix, we prove that for γ sufficiently large, the balanced steady state in the symmetric model, $\mu = 0.5$, is unstable if and only if Condition C1 holds:

$$(C1) \quad \frac{[(H^{1/(1-\psi)} + L^{1/(1-\psi)})]^{2\psi}}{(HL)^{1/(1-\psi)}} > \frac{\bar{s}f(\mathcal{K}_A)}{\nu}.$$

Condition C1 holds in reasonable settings. For example, if the $\psi = 0.5$ and skills are uniformly distributed on $[0, 1]$, then the condition requires only that

$$(19) \quad \frac{1}{H^2} + \frac{1}{L^2} > \frac{1}{\nu}$$

where we note that if L is small, then $1/L^2$ is large. Qualitatively, this condition holds when skills are difficult to acquire, and when there is sufficient dispersion in skills as captured by $f(\mathcal{K}_A)$ relative to the mean skill acquired, $\bar{s}$. If C1 does not hold, then for $\gamma > \hat{\gamma}$, there are at least five steady states and $\mu_\infty = 0.5$ is stable.⁵

We now turn to the efficiency properties of human capital investments by job applicants. We next show that job applicants make socially suboptimal human capital investments. In par-

ticular, job applicants devote too much effort toward developing skills that most managers can evaluate, reducing average worker productivity. To show this, we contrast equilibrium human capital investments with those that a social planner would make.

A. Efficiency

Consider a social planner who seeks to maximize period output given the economy-wide distribution of managerial expertise. The social planner's problem simplifies to one of choosing the skill investments $t^o(\mu_\tau) = (t_A^o(\mu_\tau), t_B^o(\mu_\tau))$ that maximize each worker type's expected productivity—reflecting that, given correct beliefs about investments by job applicants, profit-maximizing managers hire to maximize expected output. We index the hiring standards $\mathcal{K}_A^o(\mu_\tau)$ and $\mathcal{K}_B^o(\mu_\tau)$ by o to emphasize that if $t^o(\mu_\tau) \neq t^*(\mu_\tau)$, then $\mathcal{K}_j^o(\mu_\tau) \neq \mathcal{K}_j^*(\mu_\tau)$.

Total expected period output is

$$(20) \quad \begin{aligned} & \beta_A \{ \mu_\tau \mathcal{P}_A^o H t_A^\psi [e_A E[s | s \geq \mathcal{K}_A^o] \\ & + L(1 - t_A)^\psi e_B \bar{s}] \\ & + (1 - \mu_\tau) \mathcal{P}_B^o L(1 - t_B)^\psi \\ & \times [e_B E[s | s \geq \mathcal{K}_B^o] + H t_A^\psi e_A \bar{s}] \} \\ & + \beta_B \{ \mu_\tau \mathcal{P}_A^o L(1 - t_B)^\psi [e_A E[s | s \geq \mathcal{K}_A^o] \\ & + H t_B^\psi e_B \bar{s}] + (1 - \mu_\tau) \\ & \times \mathcal{P}_B^o H t_B^\psi [e_B E[s | s \geq \mathcal{K}_B^o] \\ & + L(1 - t_B)^\psi e_A \bar{s}] \}, \end{aligned}$$

where we omit the dependence of variables on μ_τ . The first two lines represent the expected output of type- A workers, where we integrate over the expertise of the managers who evaluate them, and the last two lines represent the expected output of type- B workers. Importantly, the social planner, unlike agents, weights the investment in the skill that the manager who evaluates the worker does not observe. It follows that agents make inefficient human capital investments in equilibrium.

⁵ Extensive numerical investigation reveals that if skills are uniformly distributed, and C1 does not hold, then there are five steady states.

PROPOSITION 3: *There is no economy in which $t^*(\mu_\tau) = t^o(\mu_\tau)$: equilibrium human capital investments never maximize expected period output.*

The proof proceeds by contradiction, assuming that for some μ_τ and some parameterization of the economy, the equilibrium and socially optimal investments correspond. This then implies that the hiring standards in the two scenarios are the same. We manipulate the first-order conditions for the two optimization problems and show that if $\mathcal{K}_i^* = \mathcal{K}_i^o$, for $i = A, B$, then a type- i agent invests more in skill i than is socially optimal. This reflects that a type- i agent internalizes the fact that a manager with expertise in skill i is more likely to hire him. As a result, from a social perspective, he underinvests in the skill j that the manager does not evaluate.

Most transparently, when all managers have expertise in skill A , the social planner would choose $t_A^o(1)$ and $t_B^o(1)$ to maximize

$$(21) \quad \beta_A \mathcal{P}_A^o H t_A^\psi [e_A E[s | s \geq \mathcal{K}_A^o] + L(1 - t_A)^\psi e_B \bar{s}] \\ + \beta_B \mathcal{P}_B^o L(1 - t_B)^\psi \\ \times [e_A E[s | s \geq \mathcal{K}_A^o] + L t_B^\psi e_B \bar{s}].$$

The social planner recognizes that even though no manager can unravel whether skill B was acquired, skill B still contributes to output. As a result, the social planner has agents invest some in skill B . In sharp contrast, in an equilibrium with $\mu_\tau = 1$, agents maximize the probability of being hired by a type- A manager by investing solely in skill A .

Altering compensation so that workers with higher observed skills receive higher wages does not alter the content of Proposition 3: the inefficient investments reflect the undercompensation for the development of the unobserved skill in terms of the wage or probability of being hired. Socially, it is optimal for agents to internalize the benefits of investments in skills that are unlikely to be observed, but optimizing agents do this only partially.

Proposition 3 ensures that in any equilibrium steady state, investments never correspond to the social optimum. However, we can search for a more limited notion of efficiency. Even

though equilibrium investments are always inefficient, it might still be that the equilibrium steady-state distribution of managerial expertise is the one that maximizes expected output given that investments are chosen by agents. We now show that, generically, this is not the case.

PROPOSITION 4: *For any $(e_A - e_B)$, and any γ , there is at most one value of β_A that can support the managerial mix μ^o that generates human capital investments that maximize expected period output, given that investments are chosen by workers.*

The result follows immediately if varying γ alters the steady-state distribution of managerial expertise, as γ does not directly affect period output for a fixed μ . What the proof does is show that given $(e_A - e_B)$, there is only one value of β_A for which $\mu_\infty = \mu^T$ so that $d\mu_\infty/d\gamma = 0$. Interestingly, in a symmetric setting where $e_A = e_B$ and $\beta_A = 0.5$, $\mu_\infty = 0.5$ is a steady state for all values of γ , and in this setting, $\mu_\infty = 0.5$ maximizes expected output given that agents choose investments.

These general insights about the inefficiency of investment do not depend on how we endogenize the dynamics of evaluation expertise. Section IV highlights an additional possible dynamic source of inefficiency, showing that compensating managers with equity can *exacerbate* investment distortions, and certainly does not eliminate them.

III. Numerical Characterization

We now illustrate our results numerically. We suppose that $H = 0.6$, $L = 0.3$, and $\psi = 1/2$. We also assume that half the population has an innate ability in developing each skill, $\beta_A = \beta_B = 0.5$; and that skill A is more productive than skill B , with $e_A = 1$ and $e_B = 0.9$. There is a measure $\nu = 0.1$ of job openings, and realized skills s are distributed uniformly on $[0, 1]$. We first solve for equilibrium outcomes in our base model. We then show how results extend when we alter core assumptions.

A. Base Model

Table 1 presents equilibrium steady states as a function of the nurture parameter γ , where μ_0

TABLE 1—STEADY STATE μ_∞ AS A FUNCTION OF Γ AND INITIAL DISTRIBUTION, μ_o

γ	μ_o	μ_∞	Equilibrium outcome			Efficient outcome		
			t_A^*	t_B^*	$E[Y^*]$	t_A^o	t_B^o	$E[Y^o]$
0		0.548	0.885	0.674	0.917	0.788	0.582	0.926
0.75		0.889	0.994	0.088	0.910	0.948	0.205	0.924
0.9	< 0.276	0.043	0.028	0.998	0.818	0.174	0.950	0.841
0.9	= 0.276	0.276	0.534	0.933	0.886	0.531	0.814	0.896
0.9	> 0.276	0.963	0.999	0.011	0.897	0.970	0.113	0.915
1	< 0.317	0	0	1	0.801	0.098	0.968	0.826
1	= 0.317	0.317	0.613	0.910	0.892	0.580	0.785	0.902
1	> 0.317	1	1	0	0.889	0.979	0.069	0.908
0.463		0.705	0.958	0.410	0.925	0.878	0.419	0.932

Notes: Starred values are competitive equilibrium outcomes; the superscript (°) denotes choices made by an optimizing social planner. Y is the expected productivity of a hire. The last row shows the managerial mix, $\mu_\infty = 0.705$, that maximizes expected employee productivity in a competitive equilibrium. It is a steady state only if $\gamma = 0.463$.

is the initial fraction of type-A agents and μ_∞ is the limiting distribution. It shows that if the nurture parameter γ exceeds 0.75, there are three steady states—an unstable steady state and two stable steady states—all of which depend on γ . Table 1 also presents the human capital investments that a social planner would select, given the associated steady-state distribution of managerial expertise. Equilibrium investments are far more distorted toward the skill that most managers evaluate than is optimal from a social perspective, especially when the steady-state distribution of managerial expertise is more unbalanced.

The stable steady states feature overspecialization in the skill that most managers can evaluate, which reduces the productivity of the average employee, $E[Y^o]$, even when the efficient skill A predominates. The more skewed the distribution of expertise, the greater is the investment distortion by agents seeking jobs. In turn, when investment distortions are greater, fewer potential employees actually acquire skills, so that the productivity of new workers is lower.

The last line of Table 1 shows that the distribution of managerial skills that maximizes expected productivity is $\mu = 0.705$. This optimal mix is supported as a steady state only when $\gamma = 0.463$. Lesser values of γ have too few managers with expertise in skill A in the long run, while greater values of γ have too many type-A managers (if the steady state is unique); or, they have either too many or

grossly too few if there are multiple steady states. So, although the initial distribution of managerial expertise may be such that the evolution of managerial expertise initially approaches the optimal mix, in which case average worker quality initially improves, the process continues thereafter. Generically, in the long run, both managerial expertise and worker investments become overly specialized.

Figure 1 illustrates the evolution of the proportion of type-A managers given nature ($\gamma = 0$) and nurture ($\gamma = 1$). All steady states occur where the $\mu_{\tau+1}(\gamma)$ curves intersect the 45° line. The curve $\mu_{\tau+1}(\gamma = 1)$ intersects the 45° line at $\mu = 0, 0.317$, and $\mu = 1$. If managerial expertise is determined solely by nurture and $\mu_o < 0.317$, then eventually all managers are of type B, even though skill B is less productive. If $\gamma = 0$, the unique steady state occurs at $\mu = 0.548$. The two curves— $\mu_{\tau+1}(\gamma = 0)$ and $\mu_{\tau+1}(\gamma = 1)$ —intersect at $\mu^T = 0.448$. At μ^T , the measure of type-A agents that become type-B managers just equals the measure of type-B hires that become type-A managers. We next investigate how outcomes are affected when we alter our core assumptions.

B. Workers Compensated according to Marginal Productivity

We have assumed that all workers receive the same wage. Suppose, instead, that workers are paid their expected marginal product. Then, a type-A agent chooses $t_A(\mu_\tau)$ to maximize

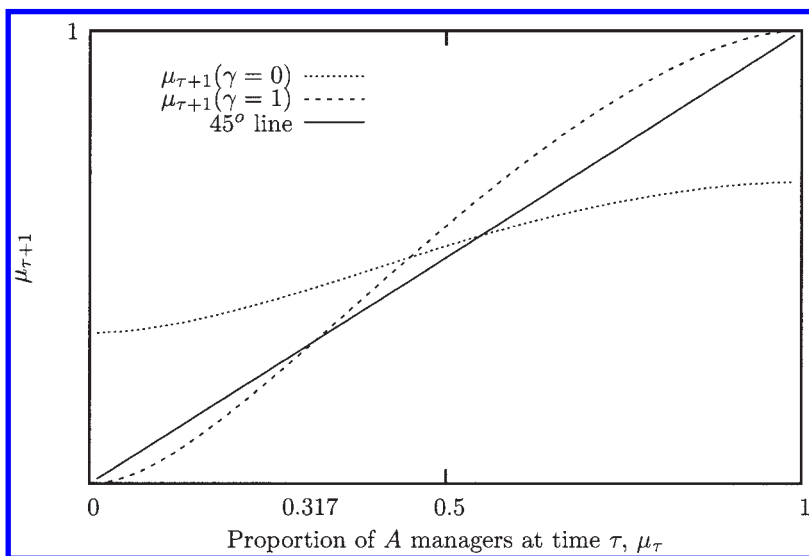


FIGURE 1. THE DYNAMIC EVOLUTION OF MANAGERIAL SKILLS

Notes: $\mu_{\tau+1}$ is plotted against μ_τ for $\gamma = 0$ and $\gamma = 1$. Steady states correspond to intersections with the 45° line. For $\gamma = 1$, if $\mu_o < (>) 0.317$, the limiting distribution consists of only type-B (type-A) managers.

$$(22) \quad \mu_\tau \mathcal{P}_A(\mu_\tau) H t_A(\mu_\tau)^\psi W_A \\ + (1 - \mu_\tau) \mathcal{P}_B(\mu_\tau) L (1 - t_A(\mu_\tau))^\psi W_B$$

where $W_j = E[e_A s_A + e_B s_B | s_j > \mathcal{K}_j(\mu_\tau)]$ is a worker's expected wage (average product) if he is hired by a type- j manager. Letting $Q = W_B/W_A$, a type-A agent's objective is

$$(23) \quad W_A [\mu_\tau \mathcal{P}_A(\mu_\tau) H t_A(\mu_\tau)^\psi \\ + (1 - \mu_\tau) Q \mathcal{P}_B(\mu_\tau) L (1 - t_A(\mu_\tau))^\psi].$$

A type-B agent has an analogous optimization problem. Hence, the optimization problems are identical to those in the base model up to the constant weighting parameter, Q . Equilibrium investments are shifted to the nonpredominant skill, reflecting that someone who acquires that skill is expected to have a higher marginal product. The shift is slight, however, as $E[e_A s_A + e_B s_B | s_A = \mathcal{K}_A(\mu_\tau)] = E[e_A s_A + e_B s_B | s_B = \mathcal{K}_B(\mu_\tau)]$, so that for reasonable parameterizations, $Q \sim 1$. The key remains that workers understand that they will not be compensated for skills that a manager cannot observe. Nu-

merical computations confirm that outcomes are almost unchanged if workers are paid their expected product rather than a flat wage: in our base-case formulation, equilibrium investments, and hence the limiting distribution of managerial skills, are shifted by less than 1 percent toward the nonpredominant skill.

C. Skills Complement Each Other in Production

Suppose now that a worker's output is $Y = e_A s_A + e_B s_B + s_A s_B$, so that skills A and B are complements in production.⁶ Hence, from a

⁶ We build in complementarities at the individual level because it is not obvious that complementarities obtain at the firm level. In white-collar jobs, the opposite seems likely: synergies of common skills are quite plausible. By way of illustration, in basketball, a lumbering big man may fit in well with a team designed for a set half-court offense but fit badly in a run-and-gun team: a team wants a common style of player. So, too, small economics departments such as those at Caltech and the University of Rochester specialize in particular fields to exploit synergies in common skills. Such synergies can rationalize Unseem's (1989) findings that firms specialize in the type of worker they hire.

TABLE 2—STEADY STATE M_{∞} AS A FUNCTION OF Γ AND INITIAL DISTRIBUTION, μ_o

γ	μ_o	μ_{∞}	Equilibrium outcome			Efficient outcome		
			t_A^*	t_B^*	$E[Y^*]$	t_A^o	t_B^o	$E[Y^o]$
0		0.536	0.867	0.709	1.006	0.709	0.576	1.031
0.75		0.854	0.985	0.189	0.974	0.863	0.337	1.010
0.9	< 0.213	0.050	0.048	0.997	0.850	0.278	0.880	0.914
0.9	= 0.213	0.213	0.424	0.956	0.952	0.473	0.779	0.983
0.9	> 0.213	0.959	0.998	0.022	0.918	0.912	0.221	0.975
1	< 0.284	0	0	1	0.801	0.192	0.914	0.877
1	= 0.284	0.284	0.568	0.924	0.973	0.536	0.736	1.001
1	> 0.284	1	1	0	0.889	0.932	0.165	0.954
0.40		0.620	0.915	0.598	1.009	0.754	0.520	1.032

Notes: In this example, output is assumed to be given by $Y = e_A s_A + e_B s_B + s_A s_B$. Efficient outcomes were computed at the steady state μ_{∞} . Parameter values are the same as in Table 1.

type- i manager's perspective, the expected productivity of a job applicant who has acquired skill i is

$$(24) \quad E[Y|s_i] = (e_i + E[s_j|s_i > 0])s_i + e_j E[s_j|s_i] \\ = \left[e_i + \left(\frac{\sigma_{ij}}{\sigma_i + \sigma_{ij}} \right) \bar{s} \right] s_i + \left(\frac{\sigma_{ij}}{\sigma_i + \sigma_{ij}} \right) e_j \bar{s},$$

where $\bar{s}$ is the unconditional expected skill level. For given hiring standards, an agent's investment problem is the same as in the base model. Complementarities, however, cause managers to place a premium on multiskilled workers. As a result, managers set relatively lower hiring standards for agents who acquire the nonpredominant skill—this reflects that agents who acquire the nonpredominant skill are more likely also to have acquired the predominant skill than the converse.

Table 2 shows how steady-state outcomes are affected when skills complement each other in production. Two central findings emerge. First, the lower standard set for the nonpredominant skill due to the complementarities causes agents to make more-balanced investments. As a result, equilibrium steady states are more balanced. The impacts on investments and steady states are small, however, less than 7 percent for *any* endogenous variable. Second, and more importantly, complementarities have *far* greater impacts on efficient investments than on equilibrium investments. That is, while skill complementarities lead to more balanced equilibrium investments, the output consequences of

unbalanced investments rise by even more, so that *effective* investment distortions rise sharply. This is because complementarities cause the social planner to place a greater weight on acquiring both skills, but agents do not internalize human capital investments that are not observed by the managers who evaluate them.

D. Heterogeneous Ability/Correlated Skill Acquisition

The ability to acquire different skills may be correlated due to general ability. To capture this possibility, we assume that a type i agent with general ability ξ who makes investments $\{t'_A(\mu_\tau), t'_B(\mu_\tau)\}$ acquires skill j with probability

$$(25) \quad P_j(t'(\mu_\tau)) = \begin{cases} \xi H(t'_j(\mu_\tau))^\psi & \text{if } j = i; \\ \xi L(t'_j(\mu_\tau))^\psi & \text{if } j \neq i. \end{cases}$$

Then, a type- A agent with ability ξ solves

$$(26) \quad \max_{t_A} \Phi_A(\cdot) = \mu_\tau \mathcal{P}_A(\mu_\tau) \xi H(t_A(\mu_\tau))^\psi \\ + (1 - \mu_\tau) \mathcal{P}_B(\mu_\tau) \xi L(1 - t_A(\mu_\tau))^\psi.$$

Because ξ drops out of the first-order conditions, it follows that, for *given* hiring standards, equilibrium investments do not vary with ξ . Further, while general ability affects a manager's assessment of the probability that an agent acquires the other skill, general ability affects hiring standards only to the extent that it alters the *relative* assessments of this probability by a manager with expertise in skill A versus one

with skill B . This suggests that introducing general ability should have minimal effects on equilibrium investments. A numerical exploration of our base-case formulation confirms this. Even when $\xi \sim U[0.5, 1.5]$, so that agents differ greatly in ex ante abilities, equilibrium steady-state distributions of managerial expertise are shifted by less than 0.015 for any value of γ .

E. Targeted Evaluations

Suppose now that before making skill investments, an agent can select the expertise of the manager who evaluates him. Then, in equilibrium, as long as both skills are comparably productive, an agent invests only in the skill in which he has an innate ability, and is evaluated by a manager who observes that skill. In this setting, consider a best-case scenario in which the economy-wide distribution of managerial expertise is such that the expected productivity of the marginal hire is the same for both types. Then, given the parameterization of our base model, the hiring cutoffs are $\mathcal{K}_A(\mu_\infty) = 0.789$, $\mathcal{K}_B(\mu_\infty) = 0.877$, and equilibrium output is 0.876. This output is *far lower* than in the steady states of the environment in which workers cannot target firms. The surprising result that improving matching between worker ability and firm managerial expertise worsens equilibrium outcomes arises because agents target potential employers too well: they overspecialize in developing the skill in which they have an innate ability. This leads to overly specialized work forces, a result that would be reinforced if there were complementarities in production that raised the value of generalists.

Targeting leads to even *worse* equilibrium outcomes if too many firms in the economy have expertise in the same skill. For example, if too many firms have expertise in skill A , there are relatively fewer applicants per job opening at these firms than at firms with expertise in skill B . As a result, the marginal worker hired by a skill- A firm is less productive than the marginal worker at a skill- B firm: in the economy, too many workers with skill A are hired.

The key points to retain about targeting are as follows. First, a limited ability to improve matching of innate ability or skill development with managerial expertise *can* lead to better investment decisions by *some* agents. Better

matching, however, reinforces investment distortions by agents whose innate ability corresponds to the predominant managerial expertise. Second, if agents can target employers *too* well, then all agents inefficiently overspecialize. Third, if firms are not completely specialized, so that some firms primarily have type- A managers and the other firms primarily have type- B managers, then equilibrium dynamics mirror those of the bad nurture equilibrium, but at the firm-type level, as each firm type exhibits increasing specialization over time.

F. Multiple Manager Evaluations

Our qualitative conclusions regarding equilibrium investments *may* be significantly affected if multiple managers with different types of expertise evaluate each job applicant. For example, professional service firms such as law firms, consultancies, and investment banks have several people evaluate job applicants. In our model, if all firms employ two managers, one with expertise in skill A , and one with expertise in skill B , to evaluate each job applicant and there is no noise in their evaluations, then agents make efficient investments. In the real world, however, evaluations are noisy, and to the extent that job applicants tend to be evaluated more by one type of manager, the forces underlying the investment distortions that we document remain. One should view our framework as a modeling device that magnifies the consequences of noisy evaluation in order to make presentation clear. In the end, it becomes an empirical question about the economic magnitudes of the forces that we document in different occupations.

IV. Strategic Firm Behavior

We have assumed that firms take the managerial expertise in their personnel departments as given, determined by past hiring decisions. But what if some firms can select the expertise of their managers? Which skill should they investigate, and how does this affect worker productivity and investment decisions? To address these questions, we return to our base model in which each job applicant is evaluated by one manager, and agents cannot target applications to firms. For simplicity, we focus on a symmet-

ric framework in which the two skills contribute equally to output, $e_A = e_B = e$, and innate ability is equally divided between skills, $\beta_A = \beta_B = 1/2$.

Firms take the distribution of human capital investments in the economy as given and do not internalize the collective consequences of their expertise choices for human capital investments. With constant returns to scale in evaluation expertise, it is optimal for a firm to select managers with expertise in the same skill. The key question becomes: when should a firm specialize in the skill that most managers at other firms also evaluate? Such specialization would further distort the distribution of managerial expertise in the economy, and hence further distort human capital investments.

First, consider a single small strategic firm so that human capital investments are not affected by its managerial expertise. As long as the firm receives more job applicants who acquire the nonpredominant skill j than there are openings, then a firm evaluating skill j with $(\sigma_j + \sigma_{ij})/\nu$ applicants per slot who acquire skill j hires the top $\nu/(\sigma_j + \sigma_{ij})$ percent. Hence, the average output per employee at the firm, if it specializes in evaluating skill j , is

$$(27) \quad E[Y|\mu_\tau = 1] = \frac{e}{\nu}(\sigma_j + \sigma_{AB}) \int_{\mathcal{K}_j(1)}^{\bar{z}} sf(s) ds + \left(\frac{\sigma_{AB}}{\sigma_j + \sigma_{AB}} \right) e\bar{s}$$

where $\mathcal{K}_j(1) = F^{-1}\{1 - [\nu/(\sigma_j + \sigma_{AB})]\}$ is the ability of the marginal hire. Let $\Delta(\mu_\tau)$ be the difference in expected output from evaluating skill A rather than skill B :

$$(28) \quad \Delta(\mu_\tau) \equiv \frac{e}{\nu} \left((\sigma_A + \sigma_{AB}) \int_{\mathcal{K}_A(1)}^{\bar{z}} sf(s) ds - (\sigma_B + \sigma_{AB}) \int_{\mathcal{K}_B(0)}^{\bar{z}} sf(s) ds \right) + \frac{e\bar{s}[(\sigma_A + \sigma_{AB}) - (\sigma_B + \sigma_{AB})]\sigma_{AB}}{(\sigma_A + \sigma_{AB})(\sigma_B + \sigma_{AB})}.$$

The firm selects managers with expertise in skill A if and only if $\Delta(\mu_\tau) \geq 0$.

Suppose, first, that managerial expertise in the economy is approximately evenly divided between the two skills, i.e., $\mu_\tau = 0.5 + \varepsilon$, for $\varepsilon > 0$, small. Then $\sigma_A \sim \sigma_B \sim \sigma$, $\mathcal{K}_A \sim \mathcal{K}_B \sim \mathcal{K}$ and

$$(29) \quad \Delta'(0.5) = e \frac{(\sigma'_A - \sigma'_B)}{\sigma_A + \sigma_{AB}} \times \left[E[s|s \geq \mathcal{K}] - \mathcal{K} - \frac{\sigma_{AB}}{\sigma_A + \sigma_{AB}} \bar{s} \right].$$

Because $\sigma'_A > 0 > \sigma'_B$, it follows that $\Delta'(0.5)$ takes the sign of the expression inside the square brackets, and hence so does $\Delta(\mu_\tau)$. The first term, $E[s|s \geq \mathcal{K}] - \mathcal{K}$, is the gain from seeing more skilled people: the firm replaces a hire who just meets the minimum hiring standard, $\mathcal{K}$, with someone who has the average skill of a new hire, $E[s|s \geq \mathcal{K}]$. The latter term is the opportunity cost of investigating the predominant skill A : an agent who acquires skill B is more likely to have acquired both skills, where $\sigma_{AB}/(\sigma + \sigma_{AB})$ is the probability an agent acquires the “other” skill and $\bar{s}$ is its expected value.

If there are far more skilled agents than jobs, $E[s|s \geq \mathcal{K}] \sim \mathcal{K}$, so that $\Delta'(0.5) \sim -[\sigma_{AB}/(\sigma + \sigma_{AB})]\bar{s} < 0$. Since $\Delta(0.5) = 0$, it follows that it is best to investigate the nonpredominant skill at $\mu_\tau = 0.5 + \varepsilon$. If, instead, there are only a few more skilled agents than jobs, $\mathcal{K} \sim 0$, so that $E[s|s \geq \mathcal{K}] \sim \bar{s}$. Hence $\Delta'(0.5) \sim [\sigma/(\sigma + \sigma_{AB})]\bar{s} > 0$, and the firm should investigate the predominant skill.

More generally, if there is enough dispersion in the realized skills of hires relative to the mean skill that individuals develop, then $\Delta'(0.5)$ is positive, and the firm should investigate the predominant skill. If, instead, there is less dispersion in the realized skills of hires, it is optimal to investigate the nonpredominant skill. That is, the firm should investigate the predominant skill if and only if the value of selecting among more applicants with skills exceeds the value of hiring more multiskilled applicants.

Now suppose that far more managers in the economy have expertise in skill A . If this majority is large enough, then almost no agent

TABLE 3— Δ : THE DIFFERENCE IN EXPECTED OUTPUT FROM EVALUATING SKILL A RATHER THAN B

$\bar{y}$	μ_τ : Proportion of managers skilled in A					
	0.5	0.51	0.6	0.7	0.8	0.9
0.2	0	−0.0007	−0.007	−0.016	−0.023	−0.011
	(0.335,0.335)	(0.338,0.332)	(0.361,0.306)	(0.385,0.270)	(0.409,0.221)	(0.433,0.144)
	[0.072,0.640]	[0.072,0.640]	[0.071,0.645]	[0.068,0.659]	[0.062,0.684]	[0.045,0.716]
0.3	0	−0.0003	−0.003	−0.005	0.001	0.058
	(0.335,0.335)	(0.338,0.332)	(0.364,0.302)	(0.391,0.260)	(0.415,0.205)	(0.438,0.125)
	[0.072,0.640]	[0.072,0.640]	[0.071,0.646]	[0.067,0.665]	[0.059,0.692]	[0.040,0.722]
0.4	0	0.0002	0.003	0.009	0.031	0.140
	(0.335,0.335)	(0.339,0.332)	(0.366,0.299)	(0.394,0.254)	(0.419,0.195)	(0.440,0.114)
	[0.072,0.640]	[0.072,0.640]	[0.071,0.647]	[0.067,0.668]	[0.057,0.696]	[0.036,0.725]
0.5	0	0.0008	0.009	0.024	0.064	0.231
	(0.335,0.335)	(0.339,0.332)	(0.368,0.297)	(0.396,0.250)	(0.421,0.189)	(0.441,0.107)
	[0.072,0.640]	[0.072,0.640]	[0.071,0.648]	[0.066,0.670]	[0.056,0.699]	[0.034,0.727]

Notes: The table details Δ as a function of μ_τ and the dispersion in skill distribution. Below Δ are the measures acquiring each skill, (σ_A, σ_B) ; and the measures $[\sigma_{AB}, (\sigma_{AB}/\sigma_A\sigma_B)]$ in square brackets. We assume symmetry, $\beta_A = \beta_B$ and $e_A = e_B$, the probabilities of acquiring skills correspond to our base-case formulation, and realized skills are distributed uniformly on $[1 - \bar{y}, 1 + \bar{y}]$.

acquires both skills. In that case, it immediately follows that the firm should investigate the predominant skill A. More generally, $\Delta'(\mu_\tau)$ is positive if μ_τ is large enough that $\mathcal{K}_B$ is small. To see this, use the equilibrium relationship,

$$(30) \quad \mathcal{K}_A + \frac{\sigma_{AB}}{\sigma_A + \sigma_{AB}} \bar{s} = \mathcal{K}_B + \frac{\sigma_{AB}}{\sigma_B + \sigma_{AB}} \bar{s},$$

and write $\Delta'(\mu_\tau)$ as

$$(31) \quad \Delta'(\mu_\tau) = e \frac{(\sigma'_A + \sigma'_{AB})}{\sigma_A + \sigma_{AB}} \times \left[E[s|s \geq \mathcal{K}_A] - \mathcal{K}_B - \frac{\sigma_{AB}}{\sigma_B + \sigma_{AB}} \bar{s} \right] - e \frac{(\sigma'_B + \sigma'_{AB})}{\sigma_B + \sigma_{AB}} \times \left[E[s|s \geq \mathcal{K}_B] - \mathcal{K}_B - \frac{\sigma_{AB}}{\sigma_B + \sigma_{AB}} \bar{s} \right] + e \left[\frac{\sigma'_{AB}}{\sigma_A + \sigma_{AB}} - \frac{\sigma'_{AB}}{\sigma_B + \sigma_{AB}} \right] \bar{s}.$$

Each of the three terms in (31) is positive, as for $\mathcal{K}_B$ sufficiently small, we have

$$(32) \quad \left[E[s|s \geq \mathcal{K}_B] - \mathcal{K}_B - \frac{\sigma_{AB}}{\sigma_B + \sigma_{AB}} \bar{s} \right] > 0,⁷$$

and $\sigma'_{AB} < 0$, while $E[s|s \geq \mathcal{K}_A] > E[s|s \geq \mathcal{K}_B]$.

We next investigate numerically the properties of $\Delta(\mu_\tau)$ when skills are uniformly distributed on $[\bar{s} - \bar{y}, \bar{s} + \bar{y}]$, so that variations in the density do not affect conclusions. The numerical results in Table 3 confirm the intuition generated by the theoretical analysis. In particular, if workers are quite heterogeneous in ability ($\bar{y}$ large), it is always optimal to investigate the predominant skill, and the gain from doing so rises with the proportion of managers in the economy who evaluate that skill. That is, if observing the predominant skill allows the firm to distinguish between workers with very different skills, the value of doing so grows with μ_τ because workers are more likely to have acquired the predominant skill. If, instead, workers have sufficiently similar skills, then $\Delta(\mu_\tau)$ is a U-shaped function of μ_τ , decreasing initially in μ_τ , and then increasing for μ_τ sufficiently high. This, too, is consistent with the intuition generated by our theoretical analysis,

⁷ In particular, at $\mathcal{K}_B = 0$, we $[E[s|s \geq \mathcal{K}_B] - \mathcal{K}_B - (\sigma_{AB}/(\sigma_B + \sigma_{AB}))\bar{s}] = (\sigma_B/(\sigma_B + \sigma_{AB}))\bar{s} > 0$.

which found that for slight dispersion, $\Delta(0.5) = 0$, and $\Delta(\mu_\tau) < 0$ for μ_τ slightly greater than 0.5, but $\Delta(\mu_\tau) > 0$ for μ_τ sufficiently close to one. Also, consistent with intuition, raising the dispersion in realized skills shrinks the set of μ_τ for which it is ever optimal to investigate the nonpredominant skill.

This analysis considers a single small firm which designs its personnel department. The consequences if more firms do so are straightforward. Specifically, the presence of such strategic firms is welfare enhancing if and only if they investigate the nonpredominant skill that fewer workers acquire. In particular, if the current distribution of managers μ_τ is too unbalanced, optimizing firms evaluate the predominant skill. In turn, the strategic firms cause applicants to distort skill investments further toward the predominant skill, worsening long-run outcomes. Conversely, if the distribution of managerial expertise is more balanced, then optimizing firms may specialize in observing the scarcer skill, speeding convergence to the symmetric “good” investment equilibrium.

The long-run equilibrium consequences are immediate. If there is enough dispersion in realized abilities, it is always optimal to investigate the predominant skill. As a result, if γ is large enough or there are enough optimizing firms that there are multiple steady states, the extreme steady state distributions of managerial expertise are more unbalanced, with greater investment distortions.⁸

If, instead, there is little dispersion in realized abilities, it is optimal to investigate the nonpredominant skill if and only if the economy-wide distribution of managerial expertise is not too unbalanced. This can make the balanced steady state stable. In this case, whether the economy goes to a “bad” extreme steady state or the

“good” balanced steady state depends on whether the initial economy-wide distribution of expertise is close enough to being balanced. That is, if the initial economy-wide distribution of managerial expertise is sufficiently balanced, then strategic firms investigate the nonpredominant skill, providing a natural balancing force.

A. Managerial Compensation

In our model, managers receive the profit generated by their hires. Consequently, managers hire to maximize the expected period output by their work force, and do not internalize the implications for future hiring decisions. Suppose, instead, that managers sell the firm to the current young workers who will be next period’s managers. Then period- τ managers hire to maximize the sum of current period profit plus the share price, $\Pi_\tau + \delta P_{\tau+1}$, where $\delta P_{\tau+1} = \Pi_{\tau+1} + \delta P_{\tau+2}$. Since future generations of managers have the same considerations, it follows that the period- τ managers hire to maximize the discounted sum of profits—managers hire to maximize firm value. The question becomes: how does this affect the relative hiring standards for each skill, and hence human capital investments?

Our analysis of the optimal personnel department helps answer this question. That analysis can be used to characterize when it is optimal to shade hiring of workers toward the skill that more workers develop, which would worsen investment distortions; and when it is optimal to shade hiring toward the skill that fewer workers develop.

Most obviously, suppose that the distribution of realized skills that workers acquire is always sufficiently dispersed relative to the mean skill acquired. Then, in any future period, a manager with expertise in the predominant skill always contributes more to period profits—from an evaluation standpoint it would be optimal to have managers with expertise only in the predominant skill. As a result, managers compensated with equity favor workers who are likely to have expertise in the predominant skill, sacrificing worker productivity for evaluation gains—the expected output of the marginal predominant skill hire is less than that of his nonpredominant skill counterpart. Managers shade standards for workers with the predominant

⁸ This result is reinforced if the number of strategic firms is large enough: a coordination equilibrium results in which one designing firm’s choice of expertise depends on the expertise choices of other designing firms. For example, if all firms are strategic, there are two bad equilibria in which all firms specialize in one managerial type, and one where the economy-wide mix of expertise is evenly divided, leaving firms indifferent to the skill they evaluate. In extreme coordination equilibria, no firm wants to deviate and evaluate the “other” skill because agents put almost all their efforts into developing the skill that almost all managers can evaluate.

skill both because they are more likely to have expertise in that skill as managers and because they, in turn, are more likely to identify similarly skilled workers in the future. In this case, compensating managers with equity directly lowers current output in the economy, and it further increases the proportion of managers with the predominant expertise in future periods, reducing future output. That is, future agents will further distort their investments toward the predominant skill, both because of the greater proportion of managers with the predominant expertise and because the required expected productivity of workers observed to have the predominant expertise is lower.

Now, suppose that the mean skill of a worker is higher relative to the dispersion in realized skills. Then, in the next period, whether a manager with the predominant expertise would contribute more to profits would depend on the economy-wide mix of expertise, which determines agents' future investments. To ease discussion of the issues, suppose that the preferred expertise in the next period is the one that would contribute most to next period's profits—this would be the case if, for example, the discount factor δ is sufficiently small. Our analysis revealed that if the economy-wide distribution of managerial expertise in $\tau + 1$ is not too skewed, managers with expertise in the nonpredominant skill contribute more to profit in $\tau + 1$. As a result, managers in period τ who are compensated with equity would shade hiring standards toward workers who develop the nonpredominant skill. That is, firms give up a little in worker productivity in order to develop a personnel department that will better identify employees who will acquire both skills. In turn, compensating managers with equity causes the distribution of managerial expertise in period $\tau + 1$ to be more balanced, which then leads to more balanced future human capital investments. In the long run, the effect of compensating managers with equity can be self-enforcing: it may be that the stable steady state features a more balanced distribution of managerial expertise.

B. Affirmative Action

Consider now the implications of our results for the efficacy of temporary affirmative action policies. Suppose that both skills contribute

equally to output, but a fraction $\beta > 0.5$ of women have an innate ability to develop skill A , and the same fraction β of men have an innate ability to develop skill B . Suppose, further, that historically more men than women were employed, so that the distribution of managerial skills is skewed toward evaluating the skill B that men are more likely to develop, $\mu_\tau < 0.5$. We ask two questions. First, when can a historical bias toward hiring men become entrenched, as yesterday's male hires become today's evaluators, and male managers tend to have expertise at evaluating the skill in which men have a relative advantage? Second, when can temporary affirmative action policies that require firms to increase the number of female hires for a *finite* number of periods reverse such entrenchment in the long run?

Our analysis reveals that entrenchment can occur if nurture is a sufficient determinant of managerial expertise. So, too, a sufficiently large historical bias is entrenched if managers are compensated with equity or if enough firms strategically design their personnel departments, as such bias causes firms to emphasize the predominant skill in hiring.

Now consider the effects of temporary affirmative action plans. In our base formulation, if Condition C1 holds, affirmative action has to be targeted perfectly at the unstable steady state in order to alter long-run outcomes.⁹ If, however, managers are compensated with equity, or enough firms strategically design their personnel departments, then temporary affirmative action policies do not have to be targeted perfectly in order to have permanent welfare-enhancing impacts. A policy that mandates the increased

⁹ This discussion is not altered if managers internalize the fact that women are more likely to have innate ability at skill B and men are more likely to have innate ability at skill A . Then, a manager evaluating skill A sets a lower standard for women and a higher one for men, as a woman with skill A is more likely to have also developed skill B than is a man with skill A . The fewer managers evaluating skill B do the opposite. As a result, women invest more in skill A and men invest more in skill B . The standard difference leads to more women being hired, which moderates the stable steady states (if nurture is sufficiently important). However, the implications for the efficacy of temporary affirmative action do not change: if nature predominates, affirmative action is irrelevant in the long run, and if nurture predominates, affirmative action has to be targeted perfectly at the unstable steady state to matter.

hiring of women raises the measure of managers who evaluate the skills that most women develop. In turn, this more-balanced distribution of managerial expertise reduces investment distortions by both men and women. If this causes firms to emphasize the nonpredominant skill in hiring—i.e., if the mean skill level acquired is sufficiently high relative to the dispersion in realized skills—then this temporary intervention “takes.” That is, the endogenous response to the affirmative action policy leads to an even more balanced distribution of managerial expertise and, hence, investments so that the affirmative action policy becomes self-enforcing.

V. Conclusion

This paper derives how human capital investments are affected when managers can evaluate some attributes of job applicants more easily than others. We show that relative to the social optimum, job applicants distort investments toward skills that they expect managers to be able to evaluate. In a dynamic framework where workers become managers, we show that if managerial expertise in a skill depends to a large extent on the time invested in that skill, then managerial skills become more and more skewed over time, leading to increasingly distorted human capital investments.

Surprisingly, private efforts by firms to constitute their personnel departments can be counter-productive socially. Whenever the initial distribution of managerial expertise in the economy is sufficiently unbalanced, and worker skills are sufficiently heterogeneous, expected worker productivity in the economy is *reduced* by such efforts. This is because, in such situations, firms prefer managers who can evaluate the skill that most job applicants acquire, so that human capital investments are further distorted.

Only when the initial distribution of managerial expertise is sufficiently balanced can such private strategic behavior help aggregate outcomes. In this case, a temporary intervention to balance the distribution of managerial expertise could result in a long-run outcome with higher expected worker productivity. That is, the intervention to balance managerial expertise takes, because it is reinforced over time by the equilibrium behavior of optimizing firms.

Qualitatively, perturbations to our model that raise the weight on the predominant managerial expertise (e.g., mentoring, learning by doing) tend to give rise to more unbalanced distributions of managerial expertise and worker investments and hence reduced output in the long run. Conversely, perturbations that weight the non-predominant skill more heavily, say because workers who develop both skills can evaluate both skills, lead to more balanced distributions of managerial expertise.

Our paper can be viewed as a parable for the ever-increasing emphasis on hiring MBAs in the United States. Perhaps for some idiosyncratic reason, hiring was initially done by people in personnel departments that had few technical people. So to get hired, technically minded students acquired MBAs to develop business, rather than technical, skills. Those with business skills were hired in greater numbers and, subsequently, hiring was done by managements that were more and more specialized in business skills. This further reinforced students' incentives to pursue MBAs. In the end, technical skills were weeded out, not only from the population of workers, but also from the applicant pool, as even the technically minded ceased to acquire engineering degrees. In a more general firm hierarchy, our paper potentially explains why engineers are not promoted beyond a certain level without acquiring an MBA.

APPENDIX

We begin with a lemma that presents the continuity properties of our key equilibrium variables.

LEMMA A1: $t_A(\mu_\tau)$, $t_B(\mu_\tau)$, $\mathcal{K}_A(\mu_\tau)$ and $\mathcal{K}_B(\mu_\tau)$ are (jointly) continuous in μ . Furthermore, $\mu_{\tau+1}(\mu_\tau)$ is continuous in μ_τ .

PROOF:

Consider equations (15) and (16). Fix $\mathcal{P}_A(\mu_\tau)$ and $\mathcal{P}_B(\mu_\tau)$, so that $t_A(\mu_\tau)$ and $t_B(\mu_\tau)$ are continuous in μ_τ . Then $\mathcal{K}_A(\mu_\tau) = (1/e_A)[K(\mu_\tau) - (\sigma_{AB}/\sigma_A)e_B\bar{s}]$ and $\mathcal{K}_B(\mu_\tau) = (1/e_B)[K(\mu_\tau) - (\sigma_{AB}/\sigma_B)e_A\bar{s}]$ are

also continuous in μ_τ since they are continuous functions of $t_A(\mu_\tau)$ and $t_B(\mu_\tau)$. Hence, $\Phi_A(\mu_\tau, t(\mu_\tau))$, $\mathcal{K}(\mu_\tau)$ and $\Phi_B(\mu_\tau, t(\mu_\tau))$, $\mathcal{K}(\mu_\tau)$ are continuous in μ_τ , and $\mu_{\tau+1}(\mu_\tau)$ is also continuous in μ_τ .

PROOF OF LEMMA 1:

- (a) If all evaluators are of type j , utility maximization implies that agents invest only in skill j . Hence, if $t_A(0) = 0$ and $t_B(0) = 1$. Similarly, if $t_B(0) = 0$ and $t_A(1) = 1$.
 (b) From the first-order conditions, (13) and (14), for $0 < \mu_\tau < 1$,

$$(A1) \quad \frac{\mathcal{P}_A(\mu_\tau)}{\mathcal{P}_B(\mu_\tau)} = \frac{(1 - \mu_\tau) L t_A(\mu_\tau)^{1-\psi}}{\mu_\tau H (1 - t_A(\mu_\tau))^{1-\psi}} = \frac{(1 - \mu_\tau) H (1 - t_B(\mu_\tau))^{1-\psi}}{\mu_\tau L t_B(\mu_\tau)^{1-\psi}}$$

$$\Rightarrow \left(\frac{t_A(\mu_\tau) t_B(\mu_\tau)}{(1 - t_A(\mu_\tau))(1 - t_B(\mu_\tau))} \right)^{1-\psi} = \frac{H^2}{L^2} > 1$$

$$\Rightarrow t_A(\mu_\tau) t_B(\mu_\tau) > (1 - t_A(\mu_\tau))(1 - t_B(\mu_\tau)) \Rightarrow t_A(\mu_\tau) > 1 - t_B(\mu_\tau).$$

From part (a), we know that equality holds if μ_τ equals zero or one.

- (c) We prove part (c) by contradiction. Suppose $t_A(\mu_\tau)$ is falling in μ_τ ($\Rightarrow t_B(\mu_\tau)$ is rising in μ_τ). We demonstrate the contradiction for type- A applicants. The proof is analogous for type- B applicants. Rearranging the first-order condition (13) for $t_A(\mu_\tau)$ yields

$$(A2) \quad \mu_\tau \mathcal{P}_A(\mu_\tau) H (1 - t_A(\mu_\tau))^{1-\psi} = (1 - \mu_\tau) \mathcal{P}_B(\mu_\tau) L (t_A(\mu_\tau))^{1-\psi}.$$

On the left-hand side, μ_τ and $1 - t_A(\mu_\tau)$ both rise. On the right-hand side, $1 - \mu_\tau$ and $t_A(\mu_\tau)$ both fall. For equality to continue to hold as μ_τ rises, $\mathcal{P}_A(\mu_\tau)$ must fall and $\mathcal{P}_B(\mu_\tau)$ must rise. This follows because the number of workers hired remains constant. But if $t_A(\mu_\tau)$ falls as μ_τ rises, agent optimization implies that type- A agents spend less time developing skill A and more time developing skill B . Therefore, for a given skill realization, s , a type- A manager must set a slacker standard, because given the skill realization, the agent's expected productivity rises, as he is now more likely to have developed skill B as well. Hence, the probability a type- A manager hires rises, a contradiction.

- (d) We prove this for type- A applicants. The proof for type- B applicants is analogous. From (c), since $t_A(\mu_\tau)$ and $1 - t_B(\mu_\tau)$ increase in μ_τ , it follows that $\mathcal{P}_A(\mu_\tau)/\mathcal{P}_B(\mu_\tau)$ falls in μ_τ . However, from the first-order condition in (13), the ratio of the measures of workers hired by managers of each type is

$$(A3) \quad \frac{\mu_\tau \mathcal{P}_A(\mu_\tau)}{(1 - \mu_\tau) \mathcal{P}_B(\mu_\tau)} = \frac{L t_A(\mu_\tau)^{1-\psi}}{H (1 - t_A(\mu_\tau))^{1-\psi}},$$

which is increasing in μ_τ since $t_A(\mu_\tau)$ is increasing in μ_τ . Therefore, because the number of applicants hired is constant, the probability of being hired with skill A rises with μ_τ .

PROOF OF PROPOSITION 1:

Immediate. $\mu_{\tau+1}$ is a continuous function of μ_τ (Lemma A1), mapping $[0, 1]$ into $[0, 1]$.

PROOF OF PROPOSITION 2:

- (a) With $\beta_A = \beta_B = 0.5$ and $e_A = e_B$, when $\mu_\tau = 0.5$, $t_A = t_B$, which also implies that $\Phi_A = \Phi_B = \nu$. Then from the transition equation (9), we have $\mu_{\tau+1} = \mu_\tau = 0.5$, a steady state, independent of γ .
 (b) Suppose $\gamma = 0$. Then equation (9) becomes

$$(A4) \quad \mu_{\tau+1} = \frac{\beta_A}{\nu} \Phi_A(\cdot) = \frac{\beta_A}{\nu} [\mu_\tau H t_A(\mu_\tau)^\psi \mathcal{P}_A(\mu_\tau) + (1 - \mu_\tau) L(1 - t_A(\mu_\tau))^\psi \mathcal{P}_B(\mu_\tau)].$$

We know from (a) that $\mu_\tau = 0.5$ is a steady state. To prove uniqueness (and stability), it suffices to show that at any steady state, μ^* , $(\partial \mu_{\tau+1} / \partial \mu_\tau)|_{\mu_\tau = \mu^*} < 1$. Substituting $t_A(\mu_\tau)$ and $t_B(\mu_\tau)$ from equations (15) and (16) into equation (A4) gives

$$(A5) \quad \mu_{\tau+1} = \frac{\beta_A}{\nu} [(\mu_\tau H \mathcal{P}_A(\mu_\tau))^{1/(1-\psi)} + ((1 - \mu_\tau) L \mathcal{P}_B(\mu_\tau))^{1/(1-\psi)}]^{1-\psi}.$$

Differentiating with respect to μ_τ , and substituting for $\mu_{\tau+1}$ yields

$$(A6) \quad \begin{aligned} \frac{\partial \mu_{\tau+1}}{\partial \mu_\tau} &= \frac{\mu_{\tau+1} \left[\frac{(\mu_\tau H \mathcal{P}_A)^{1/(1-\psi)}}{\mu \mathcal{P}_A} (\mu_\tau \mathcal{P}_A)' + \frac{((1 - \mu_\tau) L \mathcal{P}_B)^{1/(1-\psi)}}{(1 - \mu) \mathcal{P}_B} ((1 - \mu_\tau) \mathcal{P}_B)' \right]}{(\mu_\tau H \mathcal{P}_A)^{1/(1-\psi)} + ((1 - \mu_\tau) L \mathcal{P}_B)^{1/(1-\psi)}} \\ &= \mu_\tau \left[t_A \frac{(\mu_\tau \mathcal{P}_A)'}{\mu_\tau \mathcal{P}_A} + (1 - t_A) \frac{[(1 - \mu_\tau) \mathcal{P}_B]'}{(1 - \mu_\tau) \mathcal{P}_B} \right] \\ &< \mu_\tau \left[t_A \frac{(\mu_\tau \mathcal{P}_A)'}{\mu_\tau \mathcal{P}_A} \right] = t_A \left[1 + \frac{\mu_\tau \mathcal{P}_A'}{\mathcal{P}_A} \right] < 1 \end{aligned}$$

where we suppress the functional dependence on μ_τ and use prime (') to represent differentiation with respect to μ_τ for clarity. The first equality follows from substitution for $t_A(\mu_\tau)$ and the fact that μ_τ is a steady state, the first inequality follows from Lemma 1(d), and the last inequality follows from $\mathcal{P}_A' < 0$ and $t_A(\mu_\tau) < 1$, for $\mu \in (0, 1)$. Hence, for $\gamma = 0$, the steady state $\mu_\tau = 0.5$ is unique.

- (c) By Lemma 1 (d), when $\gamma = 1$, then $\mu = 0$ and $\mu = 1$ are steady states. We now prove their stability. Start with equation (9):

$$(A7) \quad \mu_{\tau+1} = \frac{1}{\nu} [\beta_A(1 - \gamma + \gamma t_A) \Phi_A(\cdot) + \beta_B \gamma (1 - t_B) \Phi_B(\cdot)].$$

Differentiating with respect to μ_τ gives:

$$(A8) \quad \nu \frac{d\mu_{\tau+1}}{d\mu_\tau} = \beta_A \Phi_A t_A' + \beta_A t_A \Phi_A' - \beta_B \Phi_B t_B' + \beta_B (1 - t_B) \Phi_B'.$$

In any equilibrium, $\nu = \beta_A \Phi_A + \beta_B \Phi_B$. Differentiating with respect to μ gives: $\beta_A \Phi_A' + \beta_B \Phi_B' = 0$. So we can write (A8) as

$$(A9) \quad \nu \frac{d\mu_{\tau+1}}{d\mu_\tau} = \beta_A \Phi_A t_A' - \beta_B \Phi_B t_B' + \beta_A \Phi_A' [t_A + t_B - 1].$$

Clearly, the last term in (A9) vanishes if μ_τ approaches 0 or 1. Now, differentiating equation (13), the first-order condition for t_A with respect to μ_τ , yields

$$\begin{aligned}
 (A10) \quad & (1 - \psi)[(1 - \mu_\tau)\mathcal{P}_B L t_A^{-\psi} + \mu_\tau \mathcal{P}_A H(1 - t_A)^{-\psi}]t'_A \\
 & = L t_A^{1-\psi}[(1 - \mu_\tau)\mathcal{P}_B]' - H(1 - t_A)^{1-\psi}(\mu_\tau \mathcal{P}_A)'.
 \end{aligned}$$

But from equation (13),

$$(A11) \quad (1 - \mu)\mathcal{P}_B L t_A^{-\psi} = \mu_\tau \mathcal{P}_A H \frac{(1 - t_A)^{1-\psi}}{t_A},$$

so we can write equation (A10) as

$$(A12) \quad t'_A = \frac{t_A(1 - t_A)^\psi}{(1 - \psi)\mu_\tau \mathcal{P}_A H} [L t_A^{1-\psi}[(1 - \mu_\tau)\mathcal{P}_B]' - H(1 - t_A)^{1-\psi}(\mu_\tau \mathcal{P}_A)'],$$

which equals zero as $\mu_\tau \rightarrow 1$ (since $t_A \rightarrow 1$ as $\mu_\tau \rightarrow 1$). Similarly, $t'_B \rightarrow 0$ as $\mu_\tau \rightarrow 1$. Hence, using (A9), we have $(d\mu_{\tau+1}/d\mu_\tau)|_{\mu_\tau \rightarrow 1} = 0$. The proof that $(d\mu_{\tau+1}/d\mu_\tau)|_{\mu_\tau \rightarrow 0} = 0$ is analogous.

- (d) We now prove that there must be a $\hat{\gamma}$ such that there are multiple steady states if and only if $\gamma > \hat{\gamma}$. To see this, consider the most extreme steady states, $\mu_+(\gamma) > 1/2$ and $\mu_-(\gamma) < 1/2$, where by symmetry $\mu_-(\gamma) = 1 - \mu_+(\gamma)$. We will show that increasing γ makes these steady states more extreme. To do this, define $D(\mu_\tau, \gamma)$ to be the difference between the measure of type-B employees who become type-A managers and the measure of type-A employees who become type-B managers. That is,

$$(A13) \quad D(\mu_\tau, \gamma) \equiv \gamma[\beta_B(1 - t_B(\mu_\tau))\Phi_B(\cdot) - \beta_A(1 - t_A(\mu_\tau))\Phi_A(\cdot)].$$

A bit of algebra shows that $(\partial/\partial\mu_\tau)[(1 - t_A(\mu_\tau))\Phi_A(\cdot)] < 0$, and $(\partial/\partial\mu_\tau)[(1 - t_B(\mu_\tau))\Phi_B(\cdot)] > 0$. Hence, $D(\mu_\tau, \gamma)$ is increasing in μ_τ . Further, because when $\mu_\tau = 0$, $D(\mu_\tau, \gamma) = -\{\gamma\nu\beta_A L/[\beta_A L + (1 - \beta_A)H]\} < 0$ and when $\mu = 1$, $D(\mu_\tau, \gamma) = \{\gamma\nu(1 - \beta_A)L/[\beta_A H + (1 - \beta_A)L]\} > 0$, there exists a unique cross-over distribution of managerial skills, μ^T , such that $D(\mu^T, \gamma) \equiv 0$. In symmetric settings, $\mu^T = 1/2$.

Differentiating equation (9) with respect to γ , gives

$$(A14) \quad \frac{\partial\mu_{\tau+1}(\mu_\tau; \gamma)}{\partial\gamma} = \frac{1}{\nu} [\beta_B(1 - t_B(\mu_\tau))\Phi_B(\cdot) - \beta_A(1 - t_A(\mu_\tau))\Phi_A(\cdot)] = \frac{D(\mu_\tau, \gamma)}{\nu},$$

which is positive if $\mu_\tau > \mu^T$, and negative otherwise, and we index the transition by γ for clarity. Hence, $\mu_{\tau+1}(\mu_\tau; \gamma)$ increases in γ for $\mu_\tau > \mu^T = 0.5$; and, conversely, $\mu_{\tau+1}(\mu_\tau; \gamma)$ decreases in γ for $\mu_\tau < 1/2$. It follows that for $\tilde{\gamma} > \bar{\gamma} \geq \hat{\gamma}$, we have $\mu_{\tau+1}(\mu_+(\tilde{\gamma}); \tilde{\gamma}) > \mu_{\tau+1}(\mu_+(\bar{\gamma}); \bar{\gamma}) = \mu_+(\bar{\gamma})$. Hence, $\mu_+(\gamma)$ is increasing in γ and $\mu_{\tau+1}$ clearly crosses the 45° line from above at $\mu_+(\gamma)$, implying stability.

Skill heterogeneity. Differentiating (9) with respect to $(e_A - e_B)$ yields

$$(A15) \quad \frac{\partial\mu_{\tau+1}}{\partial(e_A - e_B)} = \beta_A \Phi'_A(1 - \gamma + \gamma(t_A - (1 - t_B))) + \gamma t'_A \beta_A \Phi_A - \gamma t'_B \beta_B \Phi_B > 0.$$

Hence, steady states are increasing in $e_A - e_B$. Next, we show that μ^T is decreasing in $e_A - e_B$. The measure of type-A managers varies with γ unless $\mu_\infty = \mu^T$ so that

$$(A16) \quad \beta_A(1 - t_A(\mu_\infty))\Phi_A(\mu_\infty) - (1 - \beta_A)(1 - t_B(\mu_\infty))\Phi_B(\mu_\infty) = 0.$$

Using the first-order conditions, (15) and (16), we rewrite (A16) as

$$(A17) \quad \mu_\infty \mathcal{P}_A[\beta_A(1 - t_A(\mu_\infty))t_A(\mu_\infty)^{\psi-1}\Phi_A - (1 - \beta_A)(1 - t_B(\mu_\infty))^\psi] = 0.$$

As $e_A - e_B$ increases, $\mathcal{P}_A$ increases and $\mathcal{P}_B$ decreases. Hence, t_A increases and t_B decreases. This implies that the term in the brackets of (A17) would be negative were μ_∞ unchanged. Hence, μ^T must fall to restore equality.

PROOF OF CONDITION C1:

In the symmetric model with $\beta_A = 1/2$ and $e_A = e_B$, at $\mu = 1/2$, $t_A = t_B = H^{1/(1-\psi)}/(H^{1/(1-\psi)} + L^{1/(1-\psi)})$, and all agents have equal chances of being hired, $\Phi_A = \Phi_B = \nu$. But since

$$(A18) \quad \Phi_A = \mu_\tau \mathcal{P}_A(\mu_\tau)H(t_A(\mu_\tau))^\psi + (1 - \mu_\tau)\mathcal{P}_B(\mu_\tau)L(1 - t_A(\mu_\tau))^\psi,$$

we can use $\Phi_A = \nu$ and $\mathcal{P}_A = \mathcal{P}_B$ when $\mu = 1/2$, to solve for $\mathcal{P}_A$ at $\mu = 1/2$:

$$(A19) \quad \mathcal{P}_A = \frac{2\nu}{[H^{1/(1-\psi)} + L^{1/(1-\psi)}]^{1-\psi}}.$$

Now, differentiating equation (9) for $\gamma = 1$ and $\beta_A = 1/2$ gives

$$(A20) \quad \begin{aligned} \mu'_{\tau+1} &= \frac{1}{2\nu} [t'_A\Phi_A - t'_B\Phi_B + t_A\Phi'_A + (1 - t_B)\Phi'_B] = \frac{1}{2\Phi_A} [2t'_A\Phi_A + (2t_A - 1)\Phi'_A] \\ &= \frac{1}{2} \left[2t'_A + (2t_A - 1) \left(\frac{\Phi'_A}{\Phi_A} \right) \right], \end{aligned}$$

using successively $\nu = \Phi_A$, and then $t_A = t_B$, $t'_A = -t'_B$, $\Phi_A = \Phi_B$ and $\Phi'_A = -\Phi'_B$, and suppressing the functions' arguments for convenience. We next simplify Φ'_A . Differentiating equation (A18) and using the envelope condition yields

$$(A21) \quad \Phi'_A = \Phi_A t_A \left[\frac{(\mu \mathcal{P}_A)'}{(\mu \mathcal{P}_A)} \right] + \Phi_A (1 - t_A) \left[\frac{((1 - \mu) \mathcal{P}_B)'}{(\mu \mathcal{P}_B)} \right] = \Phi_A \left[(2t_A - 1) \left(\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \right) \right]$$

where the second equality follows from symmetry; i.e., $((1 - \mu) \mathcal{P}_B)'/(1 - \mu) \mathcal{P}_B = -[(\mu \mathcal{P}_A)'/\mu \mathcal{P}_A]$. Substituting for Φ'_A into $\mu'_{\tau+1}$ yields

$$(A22) \quad \mu'_{\tau+1} = \frac{1}{2} \left[2t'_A + (2t_A - 1) \left(\frac{\Phi'_A}{\Phi_A} \right) \right] = \frac{1}{2} \left[2t'_A + (2t_A - 1)^2 \left(\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \right) \right].$$

From the first-order condition in (13), we obtain

$$(A23) \quad t'_A = \frac{t_A(1 - t_A)}{1 - \psi} \left[\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} - \frac{((1 - \mu) \mathcal{P}_B)'}{(1 - \mu) \mathcal{P}_B} \right] = \frac{t_A(1 - t_A)}{1 - \psi} \left[\frac{2(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \right],$$

so that

$$(A24) \quad \mu'_{\tau+1} = \left[\frac{2t_A(1-t_A)}{1-\psi} + \frac{(2t_A-1)^2}{2} \right] \left[\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \right].$$

Next, observe that because $\mathcal{P}_A = 1 - F(\mathcal{K}_A)$, we have $\mathcal{P}'_A = -f(\mathcal{K}_A)(d\mathcal{K}_A/d\mu_\tau)$, where

$$\frac{d\mathcal{K}_A}{d\mu_\tau} = -\bar{s} \frac{d}{d\mu} \left[\frac{\sigma_{AB}}{\sigma_A + \sigma_{AB}} \right].$$

Further,

$$\frac{\sigma_{AB}}{\sigma_A + \sigma_{AB}} = \frac{HL[t_A^\psi(1-t_A)^\psi + t_B^\psi(1-t_B)^\psi]}{Ht_A^\psi + L(1-t_B)^\psi}.$$

Differentiating, and exploiting symmetry, yields

$$(A25) \quad \begin{aligned} \frac{d\mathcal{K}_A}{d\mu} &= \bar{s}\psi[Ht_A^{\psi-1} + L(1-t_A)^{\psi-1}] \frac{HL2t_A^\psi(1-t_A)^\psi}{[Ht_A^\psi + L(1-t_A)^\psi]^2} t'_A \\ &= \bar{s}\psi[2Ht_A^{\psi-1}] \frac{HL2t_A^\psi(1-t_A)^\psi}{[Ht_A^\psi + (1-t_A)Ht_A^{\psi-1}]^2} t'_A = \bar{s}\psi[4t_A^{\psi-1}] \frac{Lt_A^\psi(1-t_A)^\psi}{[t_A^{\psi-1}]^2} t'_A \\ &= 4\bar{s}\psi Lt_A(1-t_A)^\psi t'_A = 4\bar{s}\psi H^{1/2} L^{1/2} t_A^{(\psi+1)/2} (1-t_A)^{(\psi+1)/2} t'_A. \end{aligned}$$

Hence, at $\mu = 1/2$,

$$(A26) \quad \begin{aligned} \frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} &= 2 + \frac{\mathcal{P}'_A}{\mathcal{P}_A} = 2 - \frac{4\bar{s}\psi H^{1/2} L^{1/2} t_A^{(\psi+1)/2} (1-t_A)^{(\psi+1)/2} f(\mathcal{K}_A) t'_A}{\mathcal{P}_A} = 2 \\ &\quad - \left(\frac{8\bar{s}\psi H^{1/2} L^{1/2} t_A^{(\psi+3)/2} (1-t_A)^{(\psi+3)/2} f(\mathcal{K}_A)}{\mathcal{P}_A(1-\psi)} \right) \frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A}. \end{aligned}$$

We can then solve for $(\mu \mathcal{P}_A)' / \mu \mathcal{P}_A$ from (A26) to get

$$\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} = \frac{2(1-\psi)\mathcal{P}_A}{(1-\psi)\mathcal{P}_A + 8\bar{s}\psi H^{1/2} L^{1/2} t_A^{(\psi+3)/2} (1-t_A)^{(\psi+3)/2} f(\mathcal{K}_A)}.$$

Plugging this back into equation (A24) gives

$$(A27) \quad \begin{aligned} \mu'_{\tau+1} &= \left[\frac{4t_A(1-t_A)}{1-\psi} + (2t_A-1)^2 \right] \left[\frac{(1-\psi)\mathcal{P}_A}{(1-\psi)\mathcal{P}_A + 8\bar{s}\psi H^{1/2} L^{1/2} t_A^{(\psi+3)/2} (1-t_A)^{(\psi+3)/2} f(\mathcal{K}_A)} \right] \\ &= \left[\frac{1-\psi(2t_A-1)^2}{1-\psi} \right] \left[\frac{(1-\psi)\mathcal{P}_A}{(1-\psi)\mathcal{P}_A + 8\bar{s}\psi H^{1/2} L^{1/2} t_A^{(\psi+3)/2} (1-t_A)^{(\psi+3)/2} f(\mathcal{K}_A)} \right] \\ &= [1-\psi(2t_A-1)^2] \left[\frac{\mathcal{P}_A}{(1-\psi)\mathcal{P}_A + 8\bar{s}\psi H^{1/2} L^{1/2} t_A^{(\psi+3)/2} (1-t_A)^{(\psi+3)/2} f(\mathcal{K}_A)} \right]. \end{aligned}$$

We can now find a condition for which $\mu'_{\tau+1}|_{\mu=1/2} > 1$. Using (A27), this requires that

$$(A28) \quad (1 - \psi)\mathcal{P}_A + 8\bar{s}\psi H^{1/2}L^{1/2}(t_A(1 - t_A))^{(\psi+3)/2}f(\mathcal{K}_A) < \mathcal{P}_A[1 - \psi(2t_A - 1)^2].$$

Subtracting $\mathcal{P}_A$ from both sides, dividing by ψ and rearranging yields

$$(A29) \quad -\mathcal{P}_A + 8\bar{s}H^{1/2}L^{1/2}(t_A(1 - t_A))^{(\psi+3)/2}f(\mathcal{K}_A) < -\mathcal{P}_A(2t_A - 1)^2 = \mathcal{P}_A(4(t_A(1 - t_A) - 1)).$$

Adding $\mathcal{P}_A$ to both sides, then substituting for $\mathcal{P}_A$ from (A19) and dividing by $8t_A(1 - t_A)$, yields

$$(A30) \quad \bar{s}H^{1/2}L^{1/2}(t_A(1 - t_A))^{(\psi+1)/2}f(\mathcal{K}_A) < \frac{\nu}{[H^{1/(1-\psi)} + L^{1/(1-\psi)}]^{1-\psi}}.$$

Substituting for $t_A(1 - t_A) = (HL)^{1/(1-\psi)}/(H^{1/(1-\psi)} + L^{1/(1-\psi)})^2$ and rearranging, we see that $\mu'_{\tau+1}|_{\mu=1/2} > 1$ if and only if

$$(A31) \quad \frac{\bar{s}f(\mathcal{K}_A)}{\nu} < H^{1/2}L^{1/2} \left(\frac{(HL)^{1/(1-\psi)}}{(H^{1/(1-\psi)} + L^{1/(1-\psi)})^2} \right)^{(\psi+1)/2} \\ \times [H^{1/(1-\psi)} + L^{1/(1-\psi)}]^{1-\psi} = \frac{[(H^{1/(1-\psi)} + L^{1/(1-\psi)})]^{2\psi}}{(HL)^{1/(1-\psi)}}.$$

If Condition C1 does not hold for $\gamma = 1$, it does not hold for any $0 < \gamma < 1$. To see this, observe from equations (9) and (A24) that the derivative of $\mu'_{\tau+1}$ with respect to γ can be written as:

$$(A32) \quad \mu'_{\tau+1} = (1 - \gamma) \left(\frac{\beta_A \Phi_A}{\nu} \left[t_A \frac{(\mu_\tau \mathcal{P}_A)'}{\mu_\tau \mathcal{P}_A} + (1 - t_A) \frac{[(1 - \mu_\tau) \mathcal{P}_B]'}{(1 - \mu_\tau) \mathcal{P}_B} \right] \right) \\ + \gamma \left(\left[\frac{2t_A(1 - t_A)}{1 - \psi} + \frac{(2t_A - 1)^2}{2} \right] \left[\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \right] \right)$$

where we have substituted

$$(A33) \quad \Phi'_A = \Phi_A \left[t_A \frac{(\mu_\tau \mathcal{P}_A)'}{\mu_\tau \mathcal{P}_A} + (1 - t_A) \frac{[(1 - \mu_\tau) \mathcal{P}_B]'}{(1 - \mu_\tau) \mathcal{P}_B} \right].$$

Exploiting symmetry, at $\mu = 1/2$, this becomes

$$(A34) \quad \mu'_{\tau+1} = \left[(1 - \gamma) \left(\frac{2t_A - 1}{2} \right) + \gamma \left(\frac{2t_A(1 - t_A)}{1 - \psi} + \frac{(2t_A - 1)^2}{2} \right) \right] \left[\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \right].$$

Differentiating with respect to γ yields:

$$(A35) \quad \frac{d(\mu'_{\tau+1})}{d\gamma} = \left[-\left(\frac{2t_A - 1}{2} \right) + \frac{2t_A(1 - t_A)}{1 - \psi} + \frac{(2t_A - 1)^2}{2} \right] \left[\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \right] \\ = [(1 - \psi)(2t_A^2 - 3t_A + 1) + 2t_A - 2t_A^2] \left[\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \left(\frac{2}{2(1 - \psi)} \right) \right] \\ = [\psi(-2t_A^2 + 3t_A - 1) + 1 - t_A] \left[\frac{(\mu \mathcal{P}_A)'}{\mu \mathcal{P}_A} \left(\frac{1}{1 - \psi} \right) \right],$$

which is positive whenever the first term is. The first term is minimized (over the range $\frac{1}{2} < t_A < 1$) at $t_A = \frac{1}{2}$, where it equals $\frac{1}{2}$. Hence, if C1 does not hold, $\mu = \frac{1}{2}$ is stable for all γ , and for $\gamma > \hat{\gamma}$, there must be at least five steady states.

PROOF OF PROPOSITION 3:

Suppose that $t^*(\mu_\tau) = t^o(\mu_\tau)$. Then $\mathcal{P}_j^o(\mu_\tau) = \mathcal{P}_j^*(\mu_\tau)$ and $\mathcal{K}_j^o(\mu_\tau) = \mathcal{K}_j^*(\mu_\tau)$, for $j = A, B$. The first-order condition characterizing the socially optimal investment by a type-A worker, $t_A^o(\mu_\tau)$, is found by differentiating equation (20) with respect to $t_A(\mu_\tau)$:

$$(A36) \quad Ht_A^{\psi-1}[\mu_\tau \mathcal{P}_A^o[e_A E[s|s \geq \mathcal{K}_A^o] + L(1 - t_A)^\psi e_B \bar{s}] + (1 - \mu_\tau) \mathcal{P}_B^o L(1 - t_A)^\psi e_A \bar{s}] \\ = L(1 - t_A)^{\psi-1}[(1 - \mu_\tau) \mathcal{P}_B^o[e_B E[s|s \geq \mathcal{K}_B^o] + Ht_A^\psi e_A \bar{s}] + \mu_\tau \mathcal{P}_A^o Ht_A^\psi e_B \bar{s}],$$

or equivalently as

$$(A37) \quad \frac{L(1 - t_A)^{\psi-1}}{Ht_A^{\psi-1}} = \frac{\mu_\tau \mathcal{P}_A^o e_A E[s|s \geq \mathcal{K}_A^o] + (\mu_\tau \mathcal{P}_A^o + (1 - \mu_\tau) \mathcal{P}_B^o) L(1 - t_A)^\psi e_B \bar{s}}{(1 - \mu_\tau) \mathcal{P}_B^o e_B E[s|s \geq \mathcal{K}_B^o] + ((1 - \mu_\tau) \mathcal{P}_B^o + \mu_\tau \mathcal{P}_A^o) Ht_A^\psi e_A \bar{s}}.$$

So, too, the first-order condition for $t_B^o(\mu_\tau)$ can be rewritten as

$$(A38) \quad \frac{Ht_B^{\psi-1}}{L(1 - t_B)^{\psi-1}} = \frac{\mu_\tau \mathcal{P}_A^o e_A E[s|s \geq \mathcal{K}_A^o] + (\mu_\tau \mathcal{P}_A^o + (1 - \mu_\tau) \mathcal{P}_B^o) Ht_B^\psi e_B \bar{s}}{(1 - \mu_\tau) \mathcal{P}_B^o e_B E[s|s \geq \mathcal{K}_B^o] + ((1 - \mu_\tau) \mathcal{P}_B^o + \mu_\tau \mathcal{P}_A^o) L(1 - t_B)^\psi e_A \bar{s}}.$$

Rewriting the first-order conditions for the privately optimal investments of agents, equations (13) and (14), so as to correspond, yields

$$(A39) \quad \frac{L(1 - t_A)^{\psi-1}}{Ht_A^{\psi-1}} = \frac{\mu_\tau \mathcal{P}_A^o}{(1 - \mu_\tau) \mathcal{P}_B^o} = \frac{Ht_B^{\psi-1}}{L(1 - t_B)^{\psi-1}}.$$

But (A39) then implies that the right-hand sides of equations (A37) and (A38) must be equal, an immediate contradiction, as $Ht_B^\psi > L(1 - t_A)^\psi$ and $Ht_A^\psi > L(1 - t_B)^\psi$.

PROOF OF PROPOSITION 4:

The measure of type-A managers varies with γ unless $\mu_\infty = \mu^T$ so that

$$(A40) \quad \beta_A(1 - t_A(\mu_\infty))\Phi_A(\mu_\infty) - (1 - \beta_A)(1 - t_B(\mu_\infty))\Phi_B(\mu_\infty) = 0.$$

Using the first-order conditions, (15) and (16), we can rewrite (A40) as

$$(A41) \quad \mu_\infty \mathcal{P}_A[\beta_A(1 - t_A(\mu_\infty))t_A(\mu_\infty)^{\psi-1}\Phi_A - (1 - \beta_A)(1 - t_B(\mu_\infty))^\psi] = 0.$$

As β_A increases, because $\mathcal{P}_A$ decreases, t_A falls and t_B rises. Hence, the term in the brackets of (A41) would be positive were μ_∞ unchanged. Hence, μ^T must rise to restore equality.

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